

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

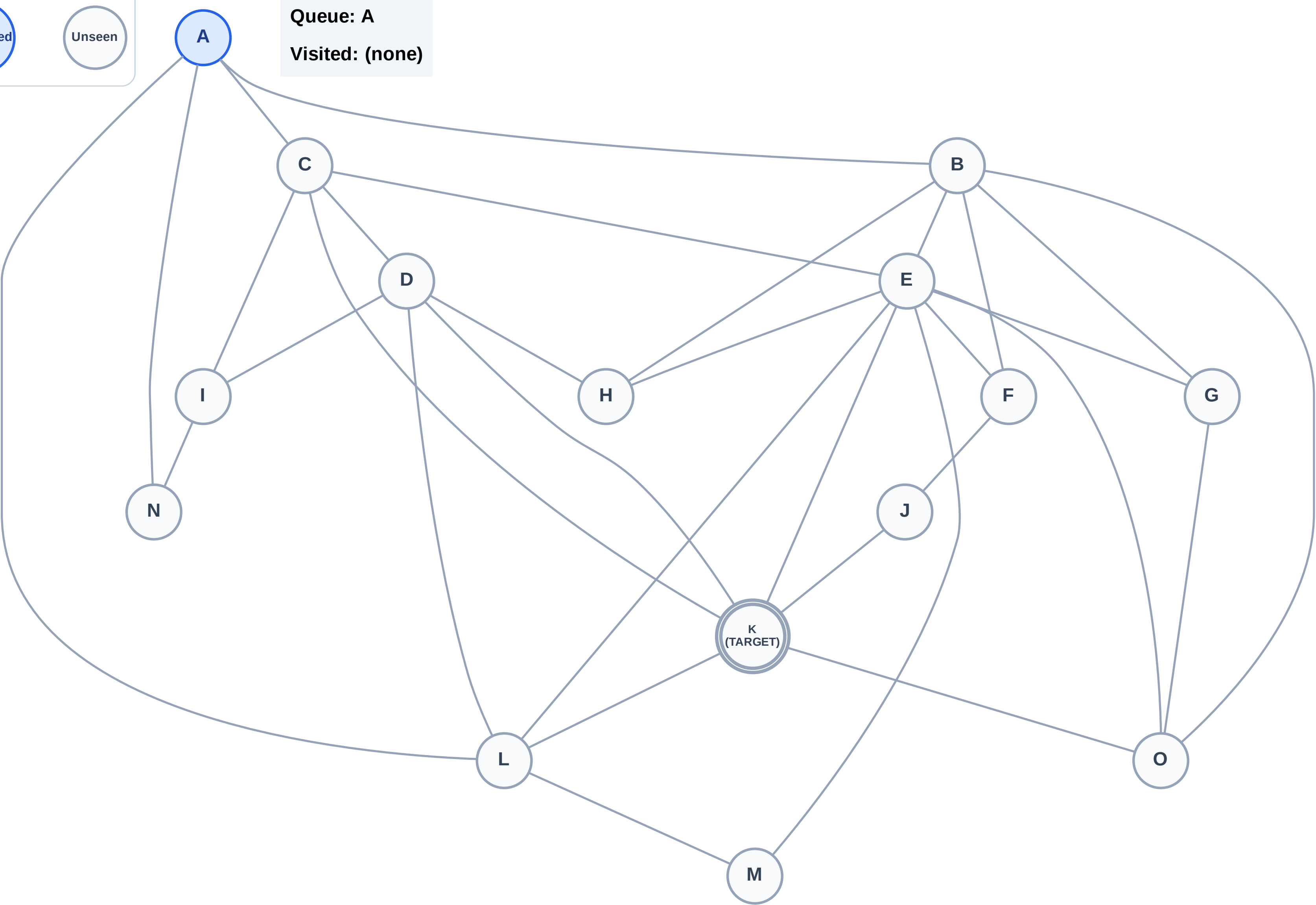
Processing

Discovered

Unseen

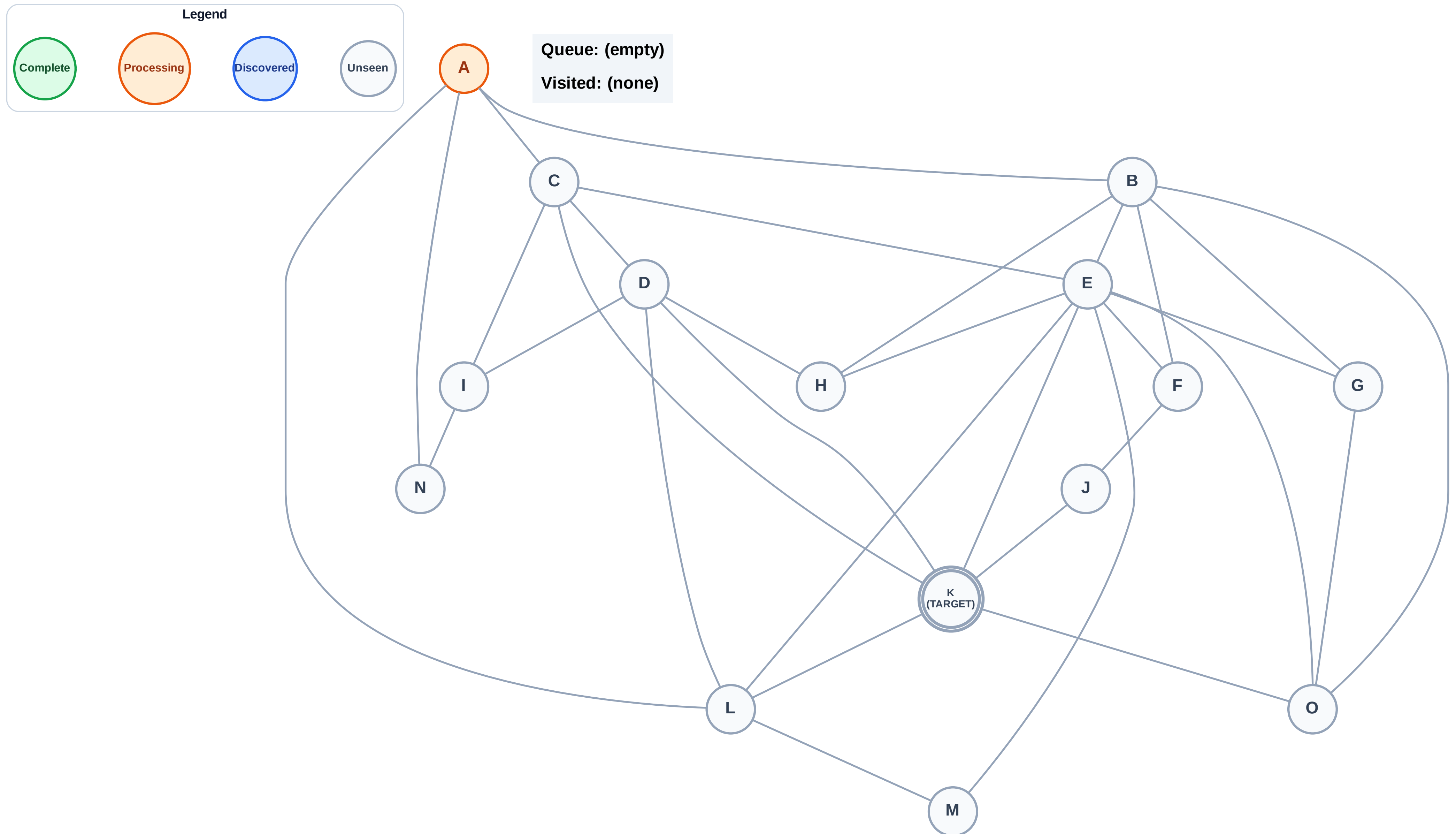
Queue: A

Visited: (none)



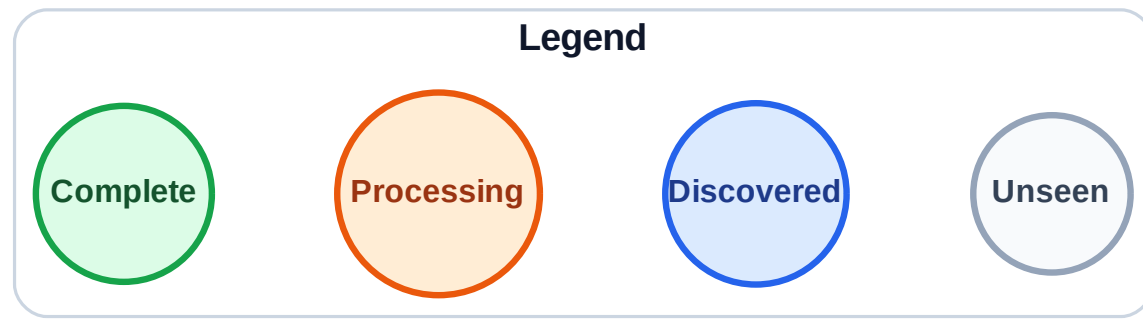
Step 2: Process node A

Step 2: Process node A

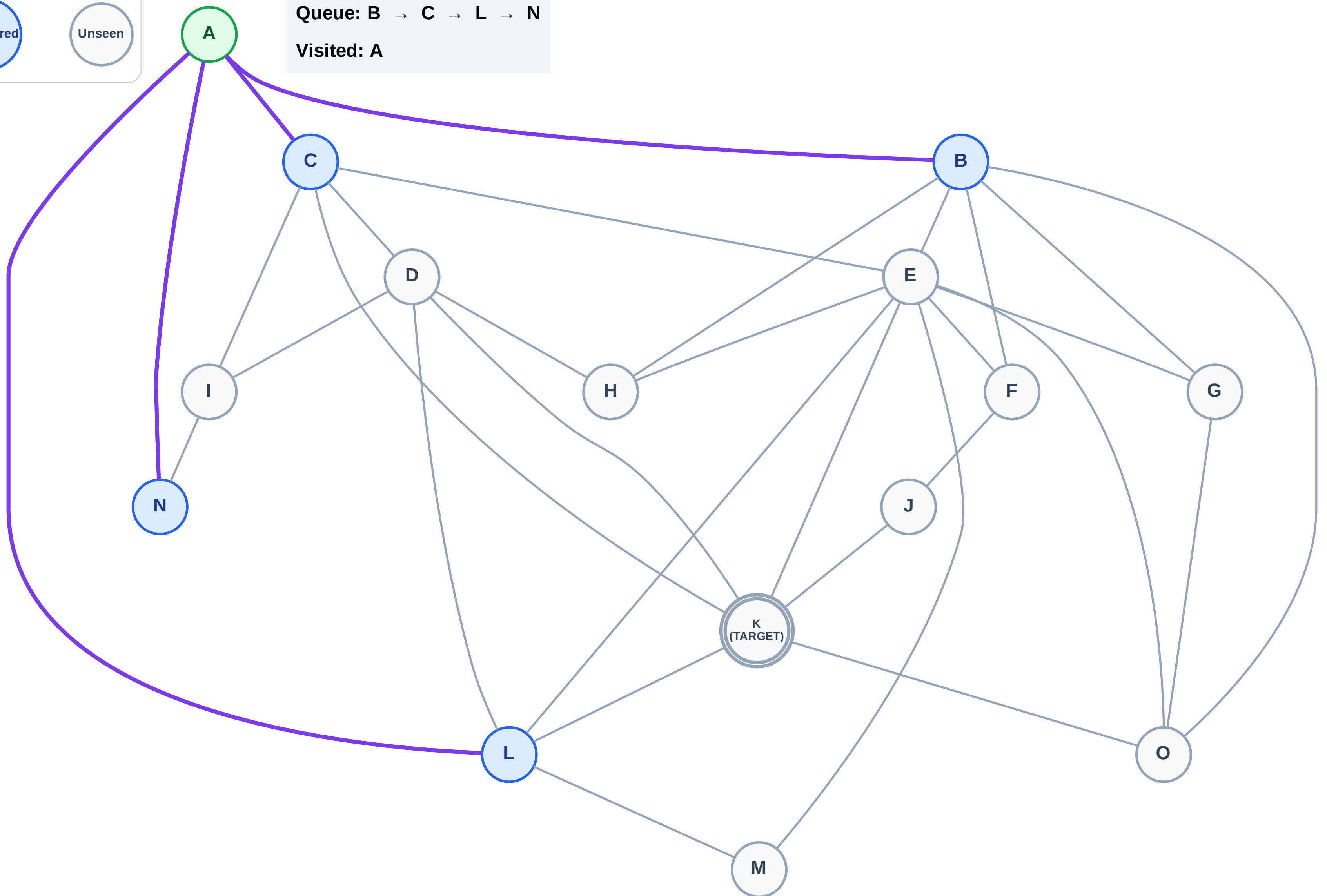


BFS Traversal

Step 3: Discover B, C, L, N from A



Queue: B → C → L → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

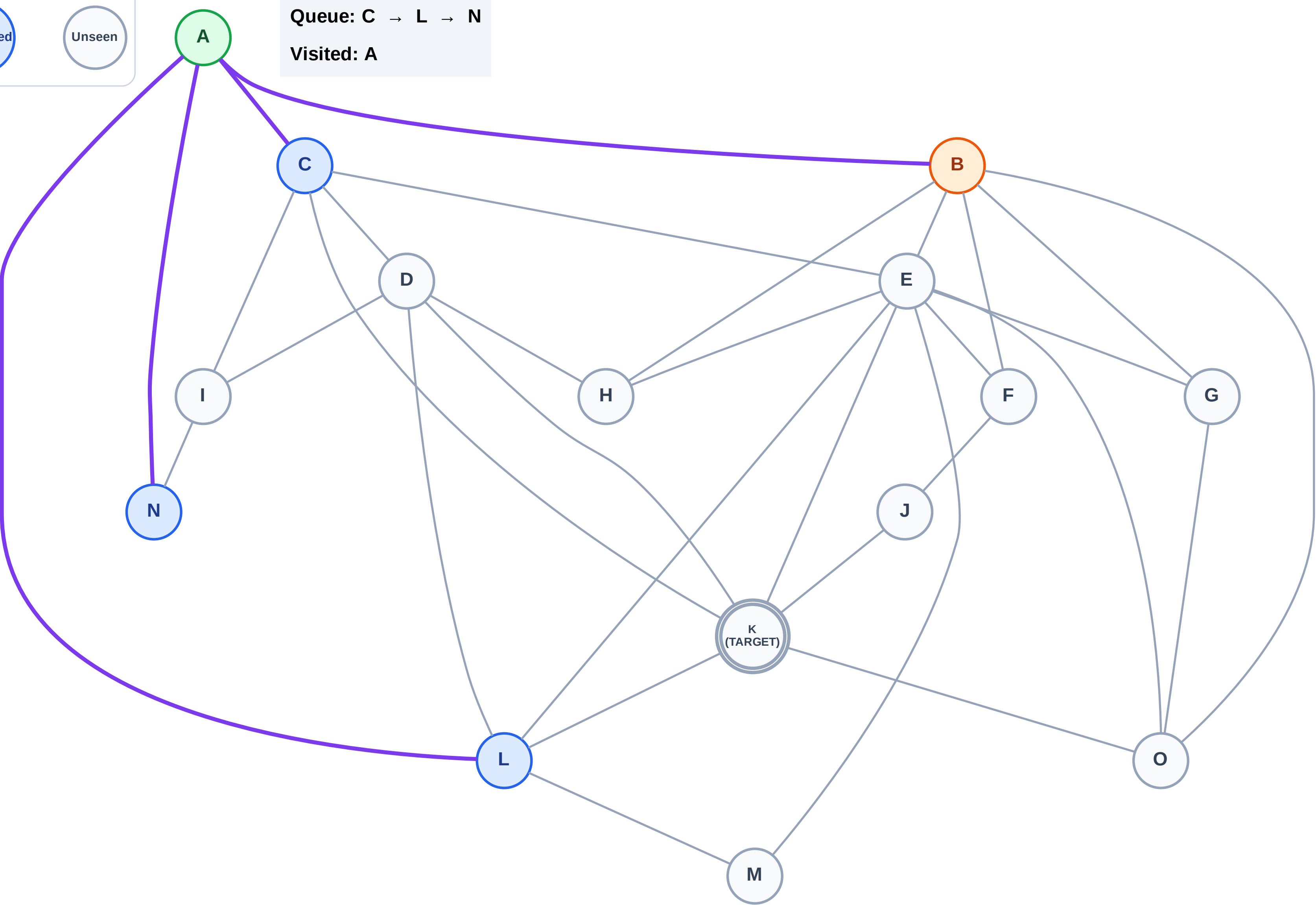
Complete

Processing

Discovered

Unseen

Queue: C → L → N
Visited: A



BFS Traversal

Step 5: Discover E, F, G, H, O from B

Legend

Complete

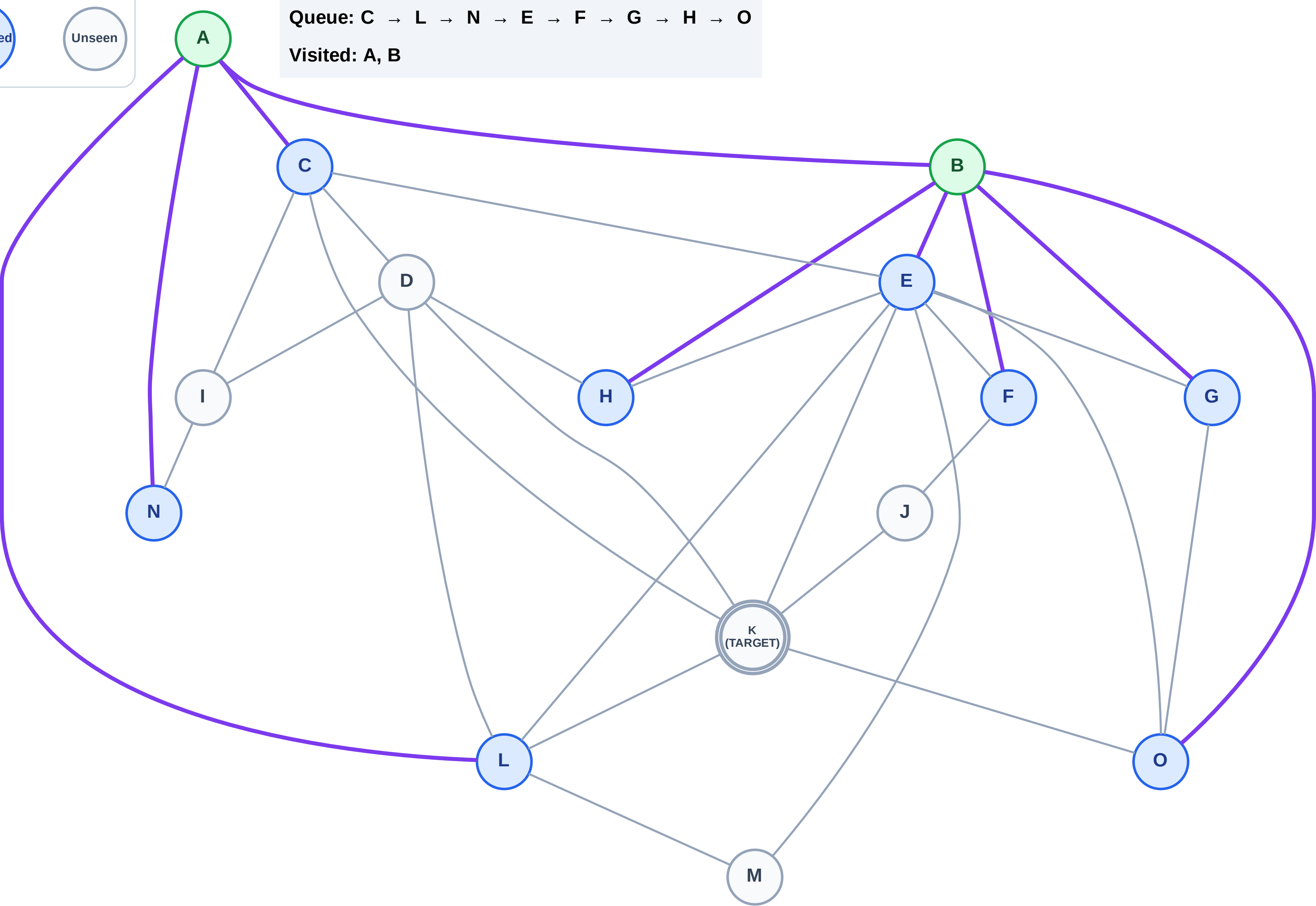
Processing

Discovered

Unseen

Queue: C → L → N → E → F → G → H → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

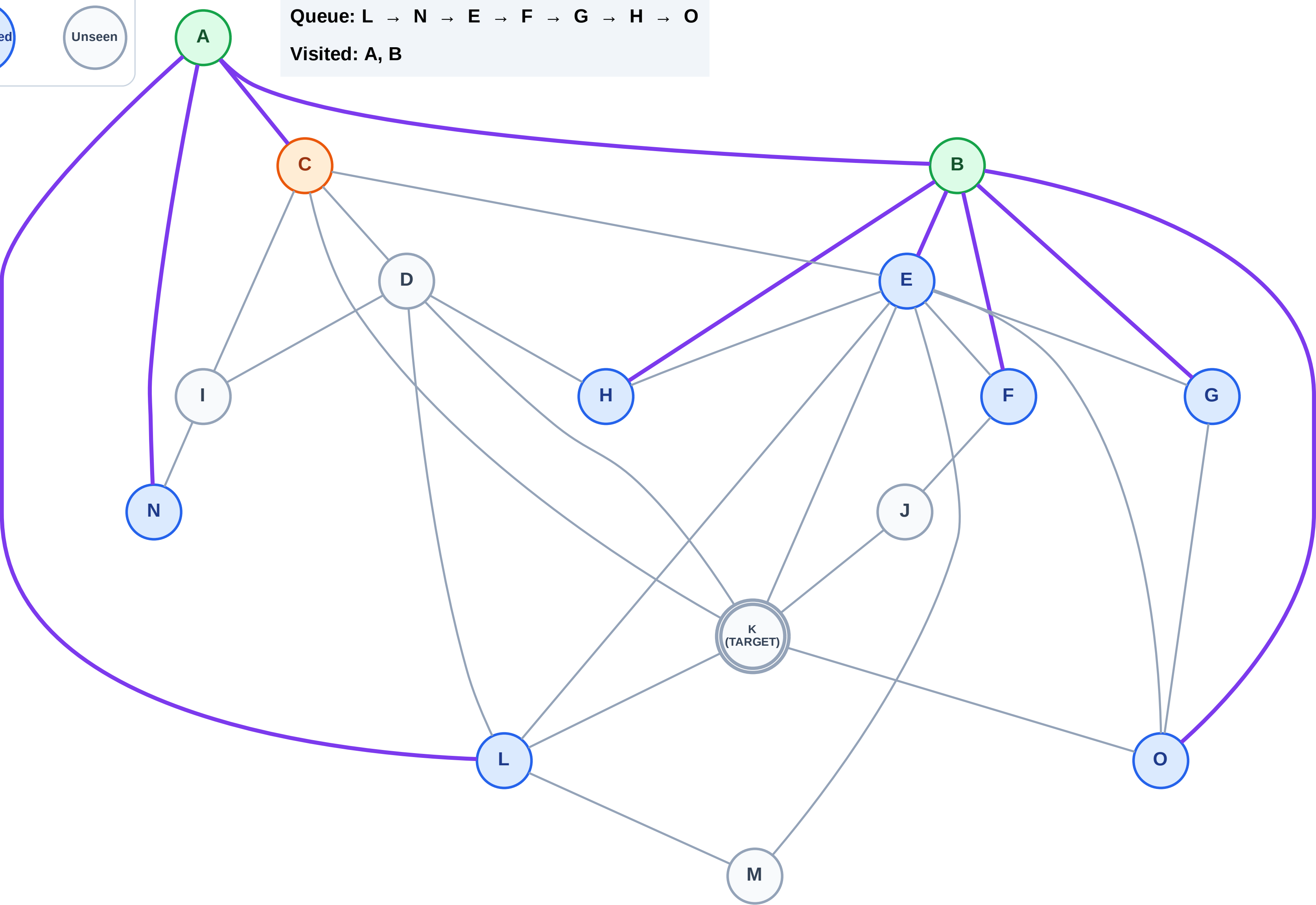
Processing

Discovered

Unseen

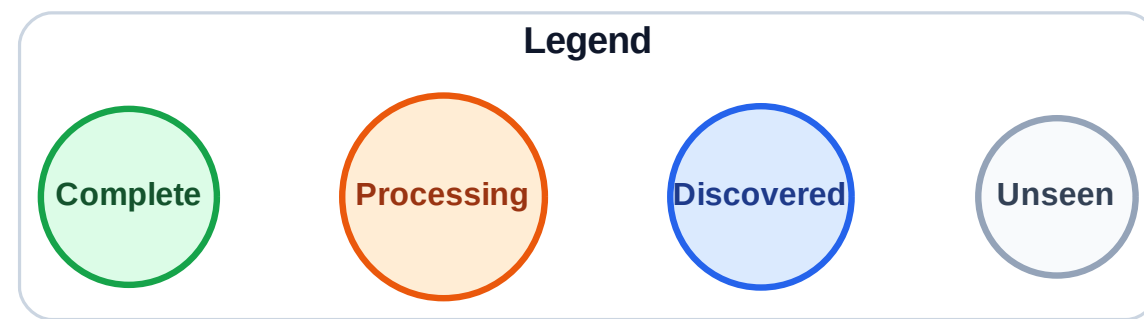
Queue: L → N → E → F → G → H → O

Visited: A, B



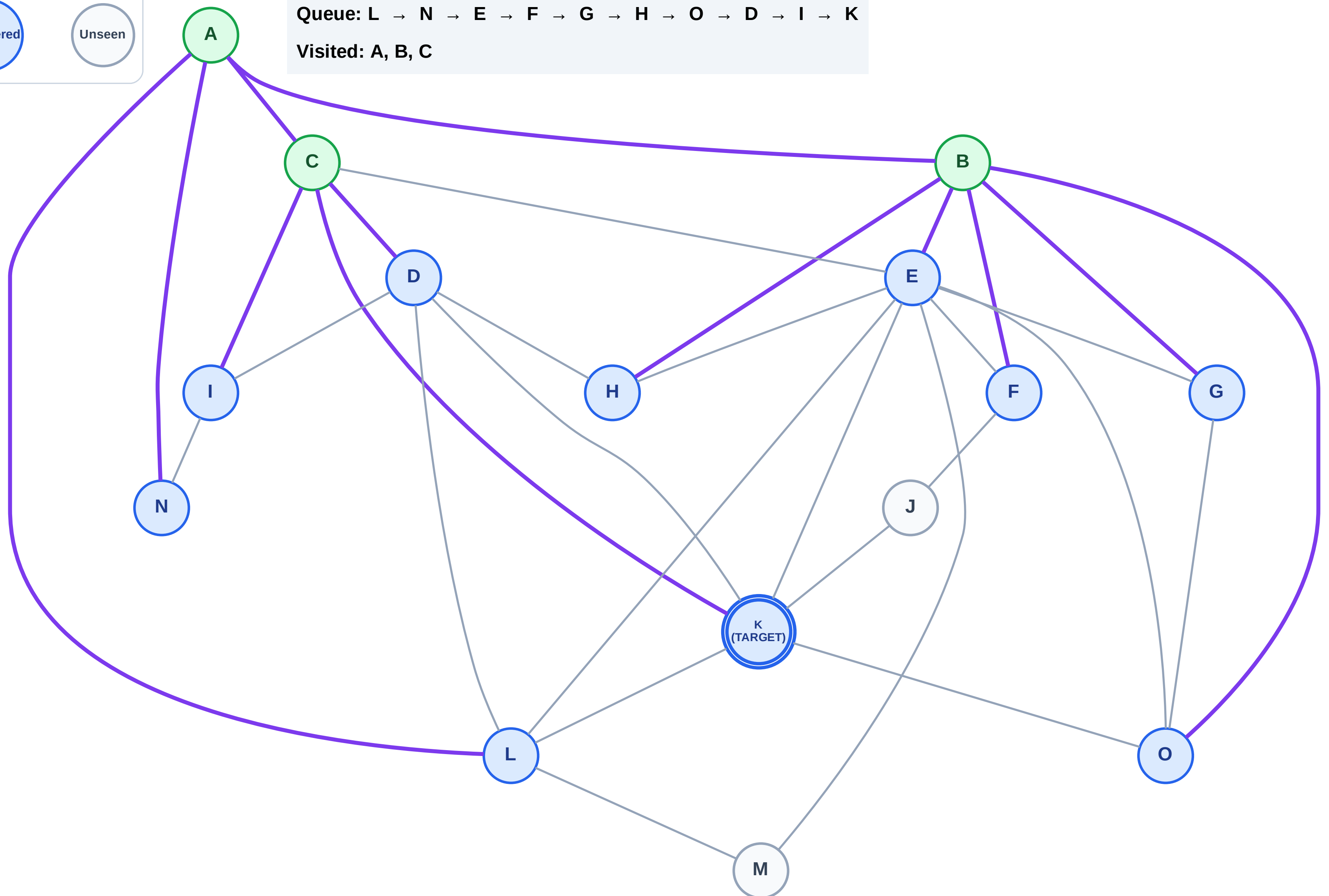
BFS Traversal

Step 7: Discover D, I, K from C



Queue: L → N → E → F → G → H → O → D → I → K

Visited: A, B, C



BFS Traversal

Step 8: Process node L

Legend

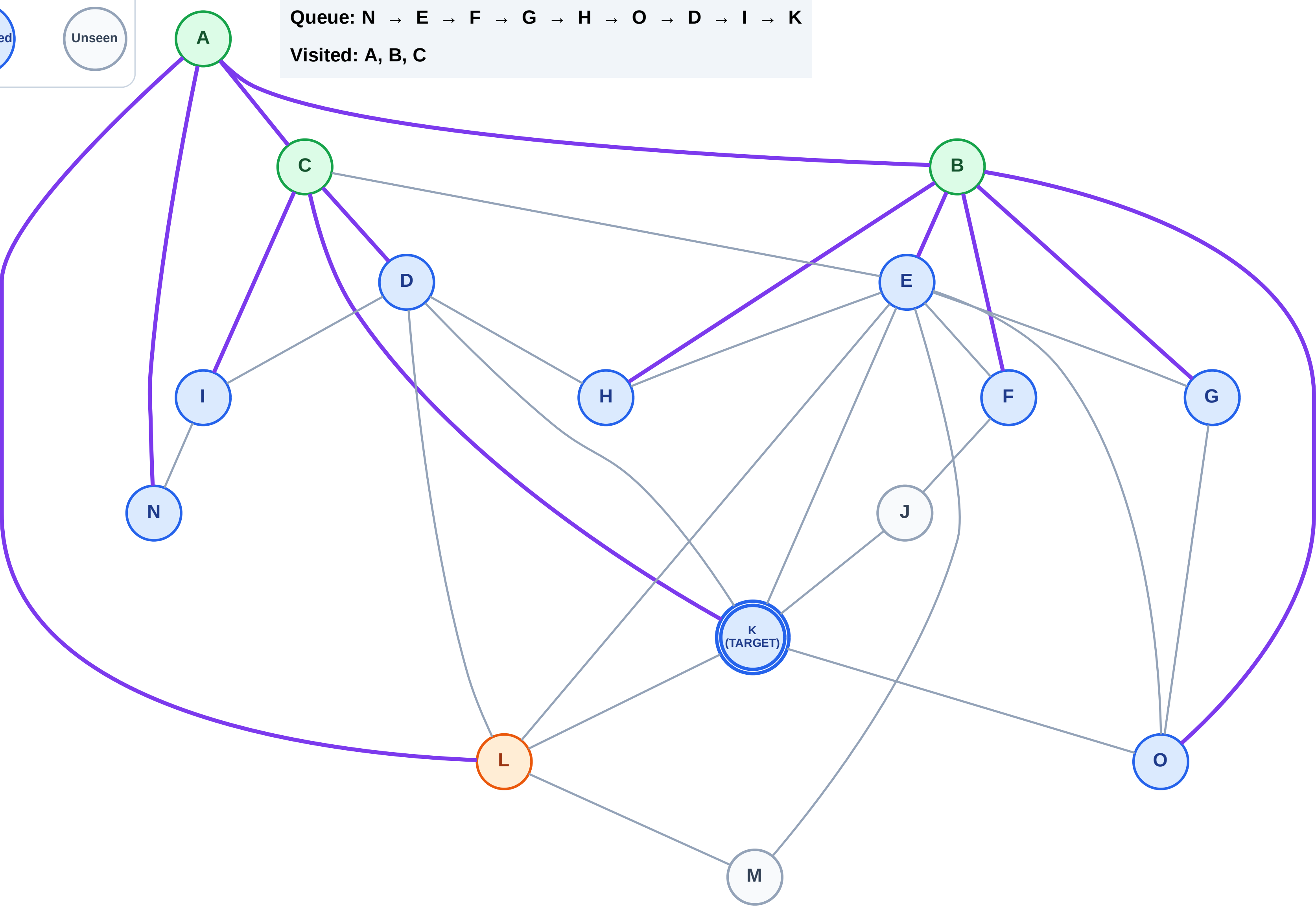
Complete

Processing

Discovered

Unseen

Queue: N → E → F → G → H → O → D → I → K
Visited: A, B, C



BFS Traversal

Step 9: Discover M from L

Legend

Complete

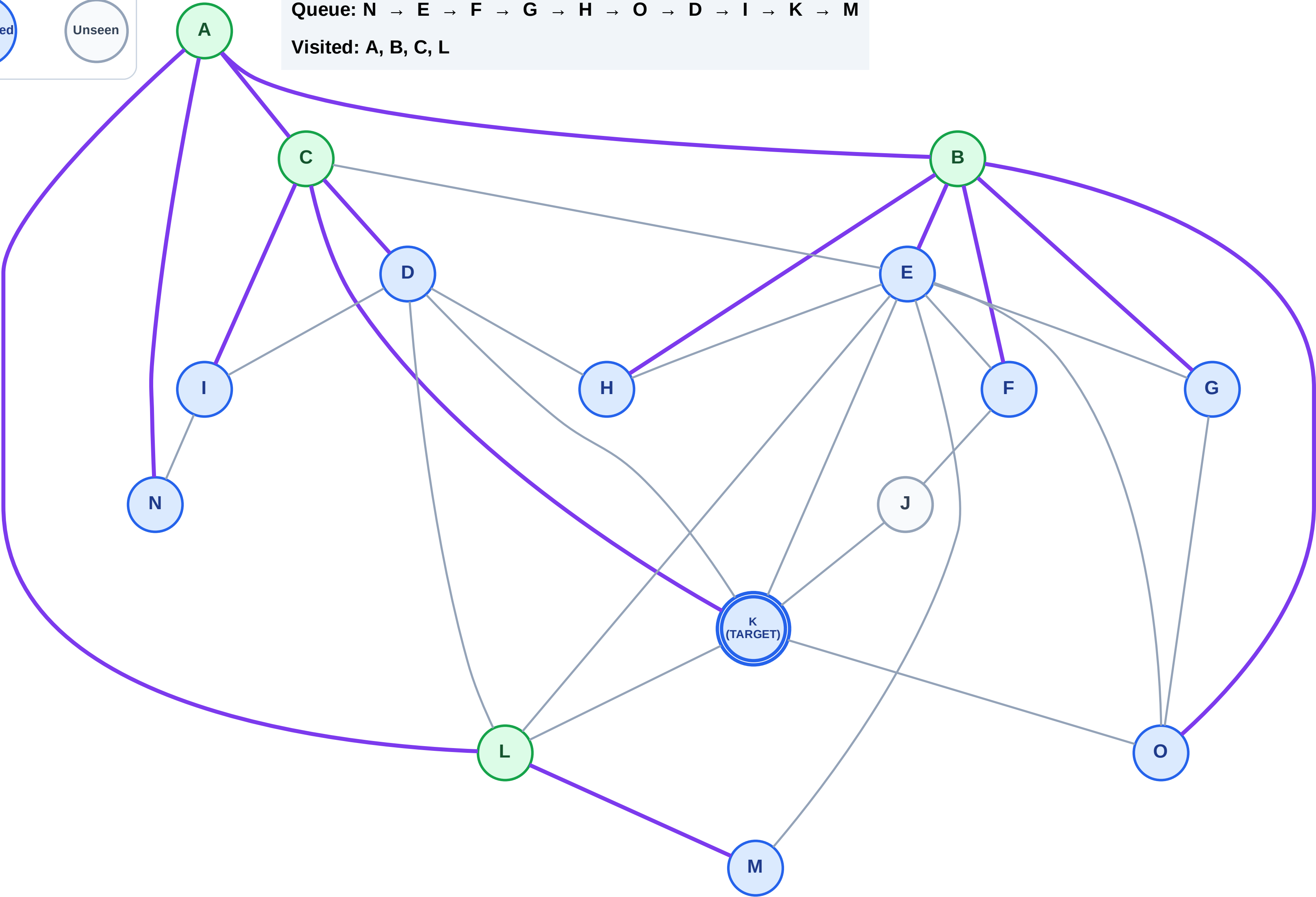
Processing

Discovered

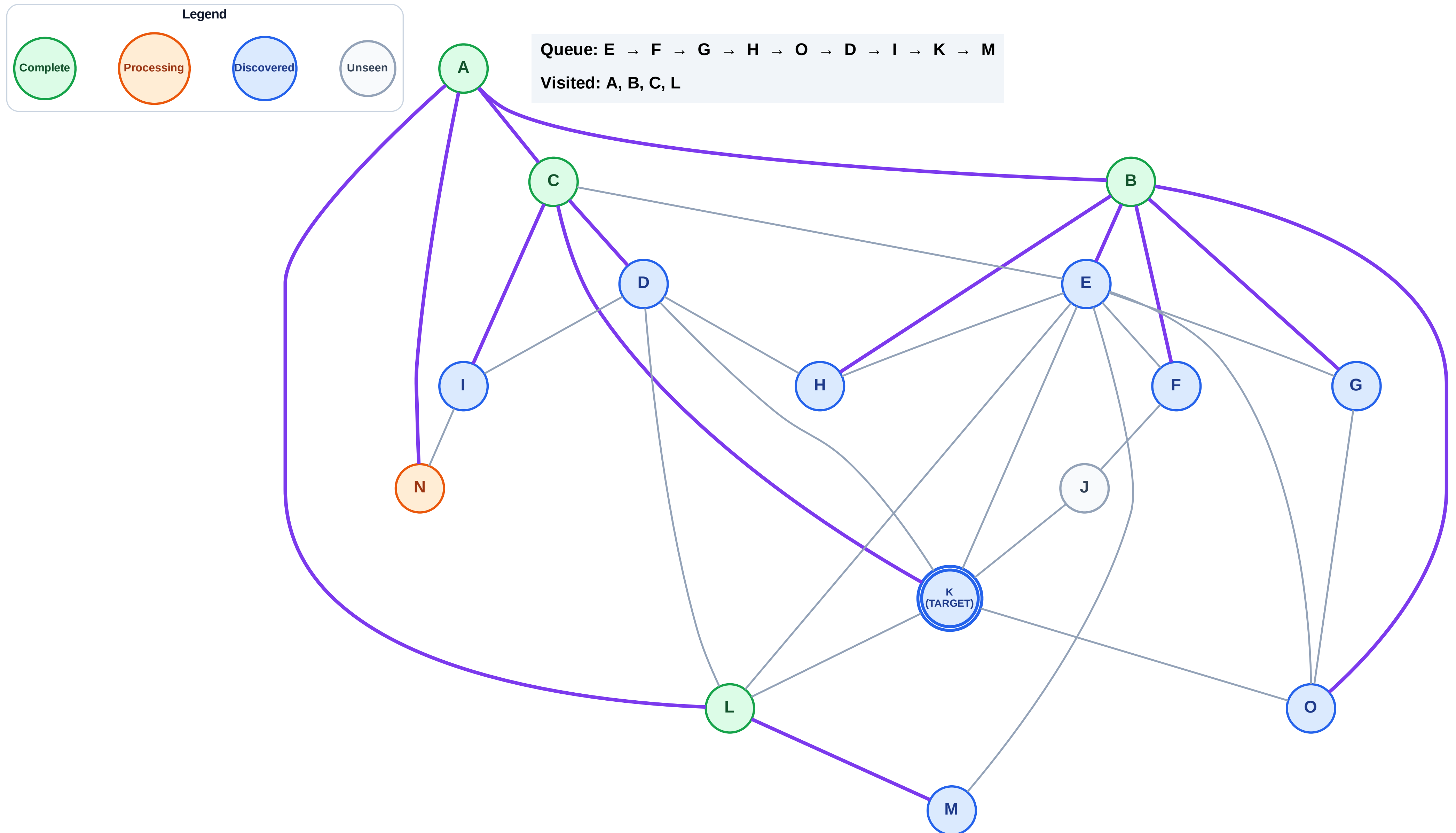
Unseen

Queue: N → E → F → G → H → O → D → I → K → M

Visited: A, B, C, L



Step 10: Process node N



BFS Traversal

Step 11: No new neighbors from N

Legend

Complete

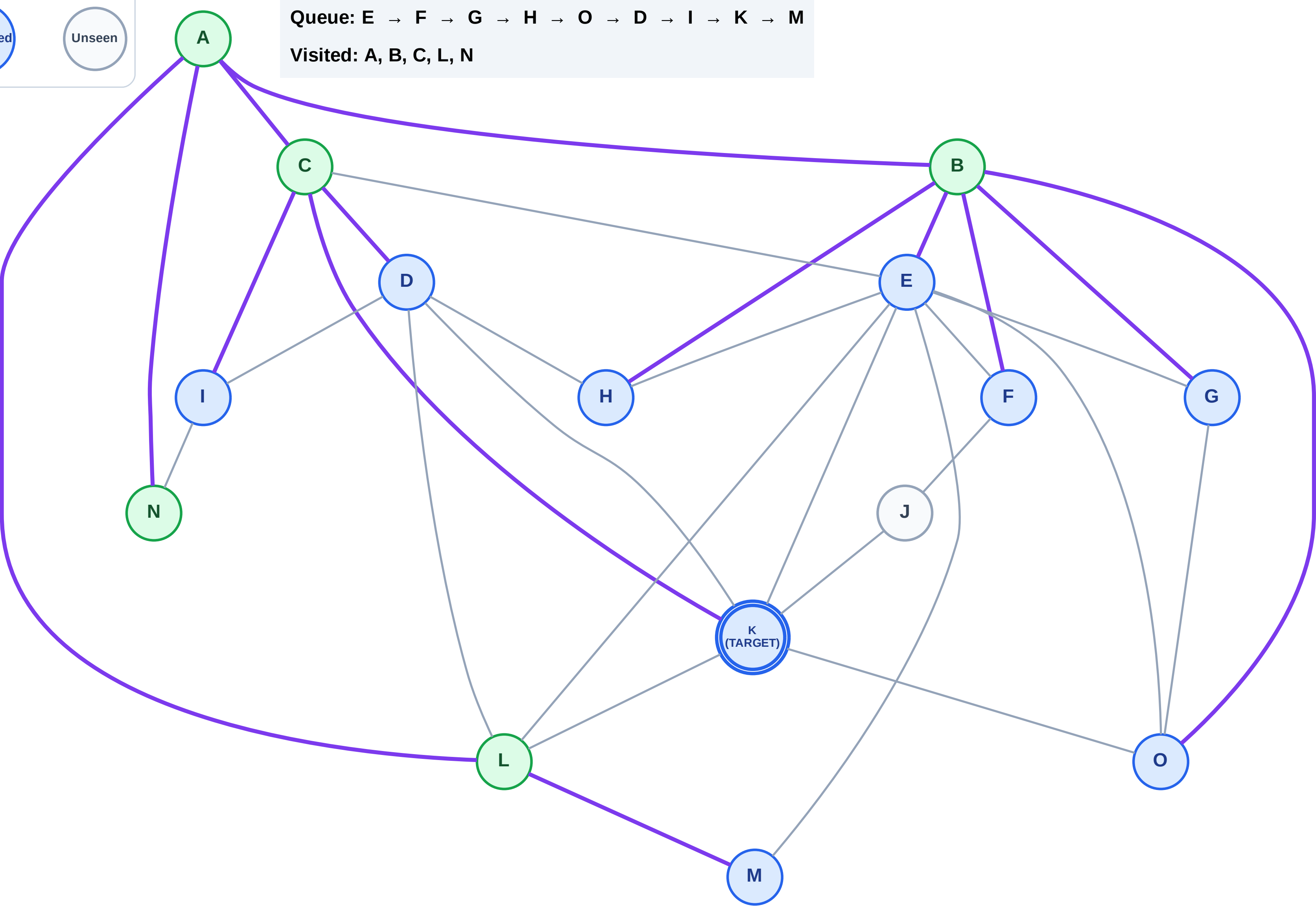
Processing

Discovered

Unseen

Queue: E → F → G → H → O → D → I → K → M

Visited: A, B, C, L, N



BFS Traversal

Step 12: Process node E

Legend

Complete

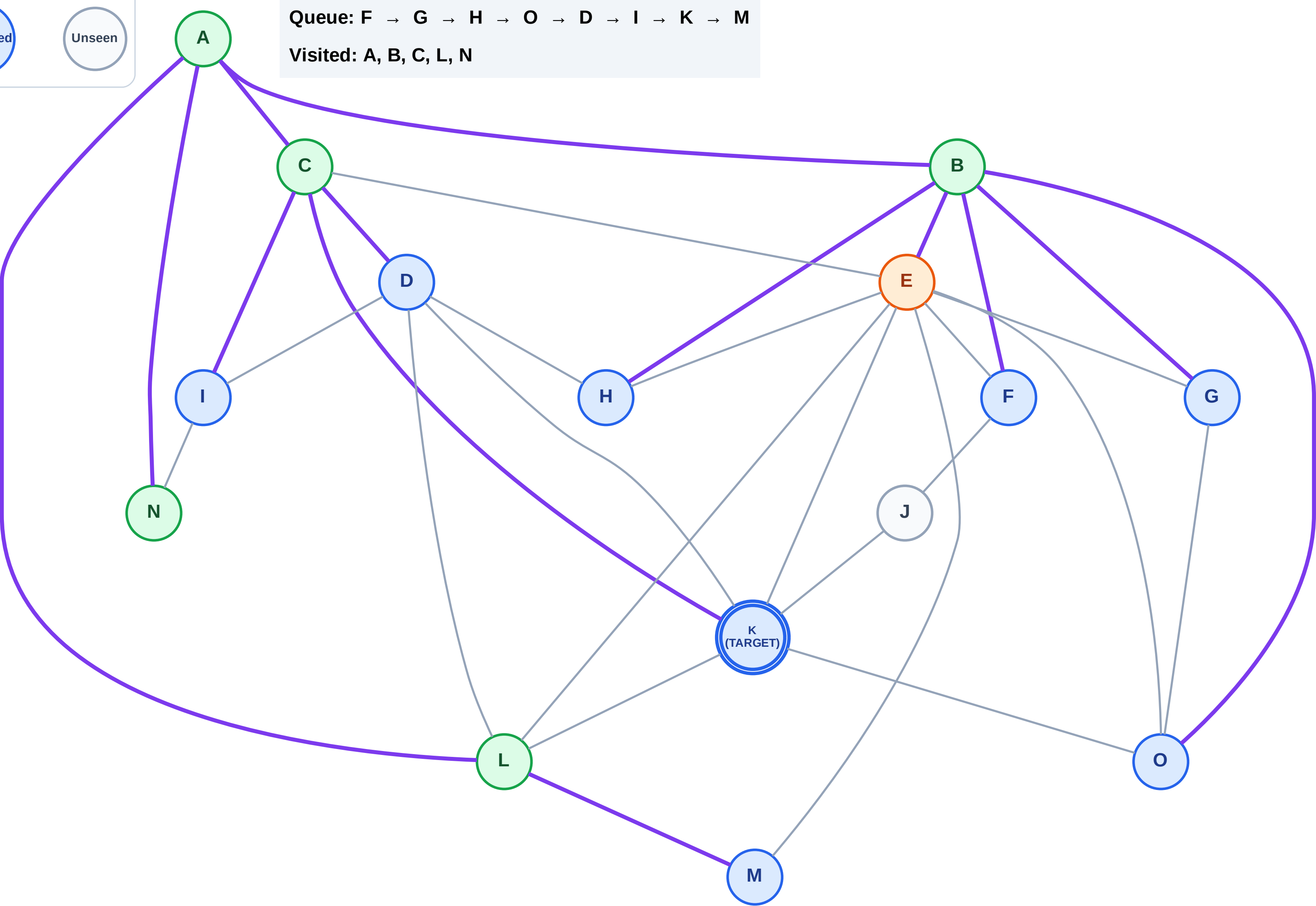
Processing

Discovered

Unseen

Queue: F → G → H → O → D → I → K → M

Visited: A, B, C, L, N



BFS Traversal

Step 13: No new neighbors from E

Legend

Complete

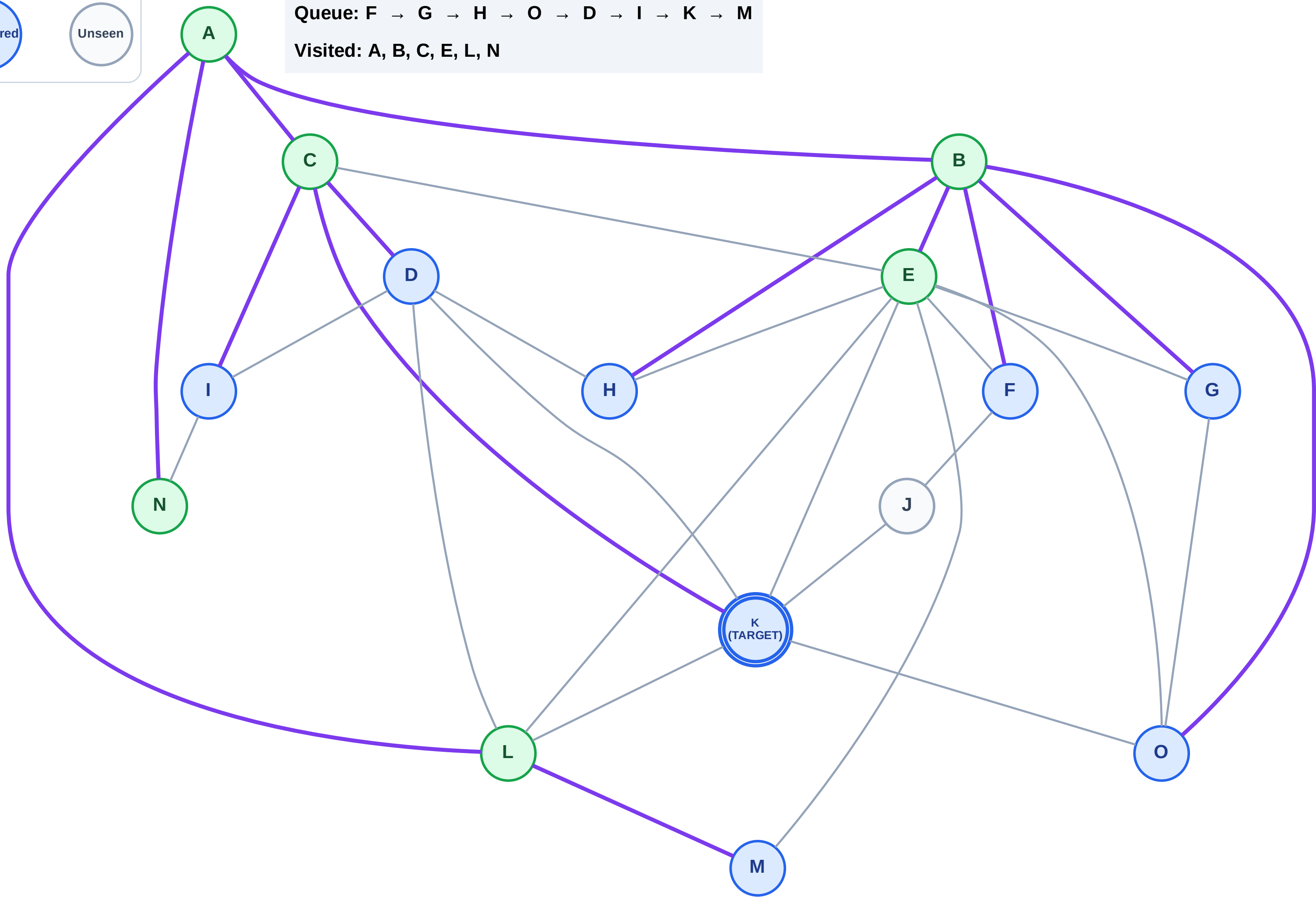
Processing

Discovered

Unseen

Queue: F → G → H → O → D → I → K → M

Visited: A, B, C, E, L, N



BFS Traversal

Step 14: Process node F

Legend

Complete

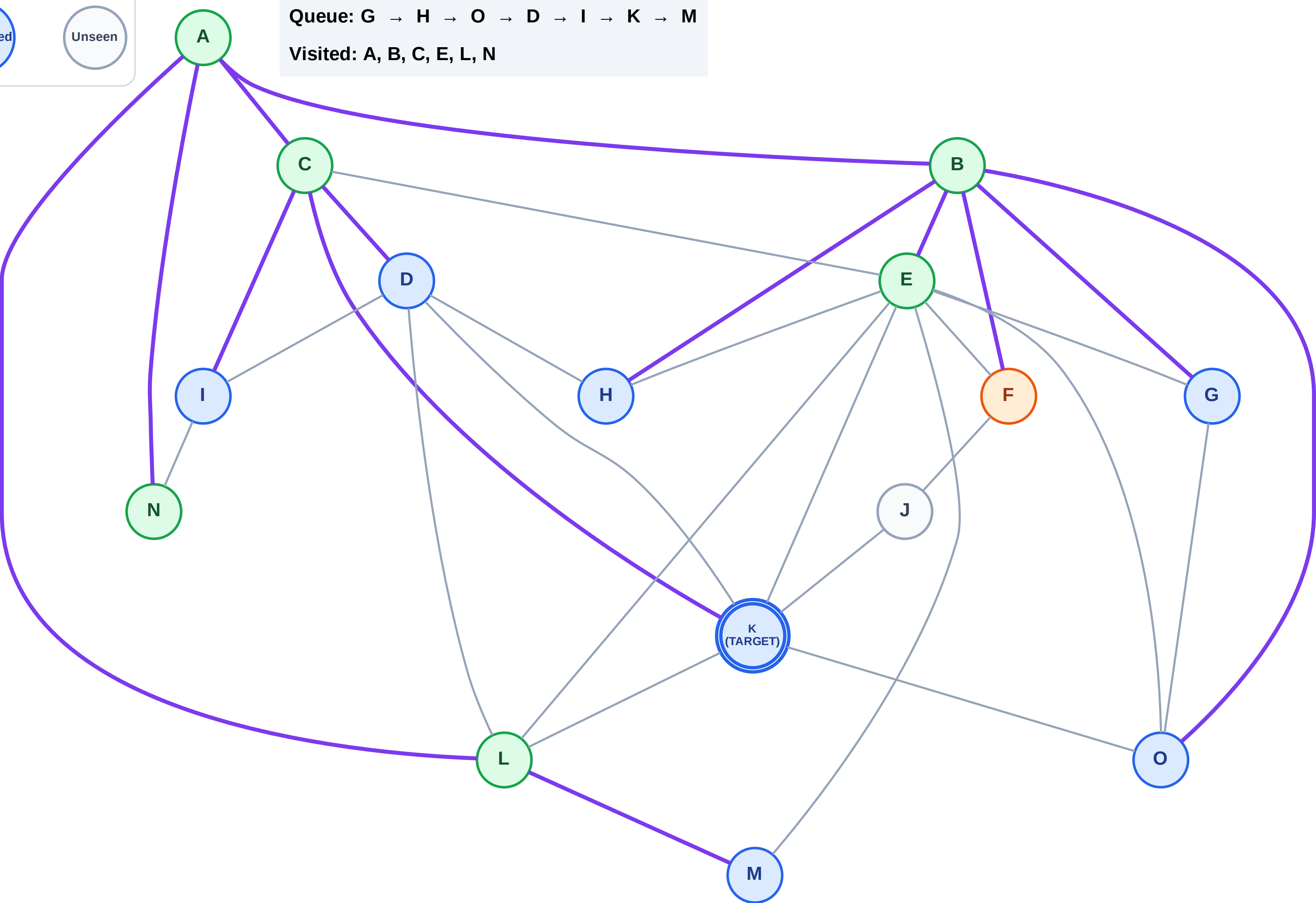
Processing

Discovered

Unseen

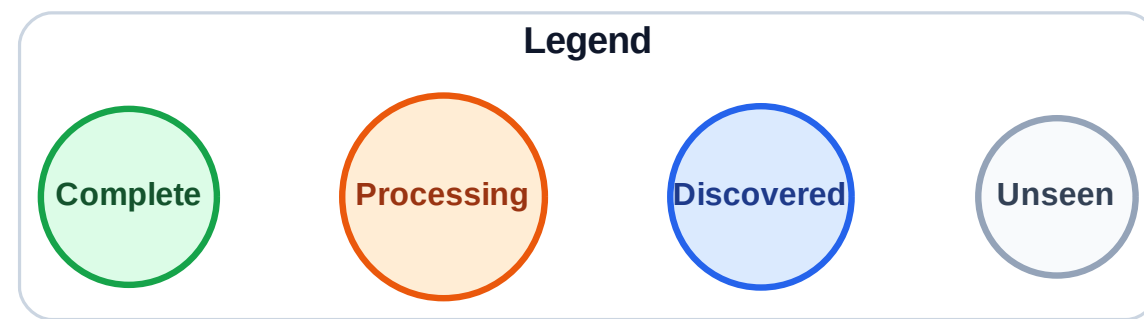
Queue: G → H → O → D → I → K → M

Visited: A, B, C, E, L, N



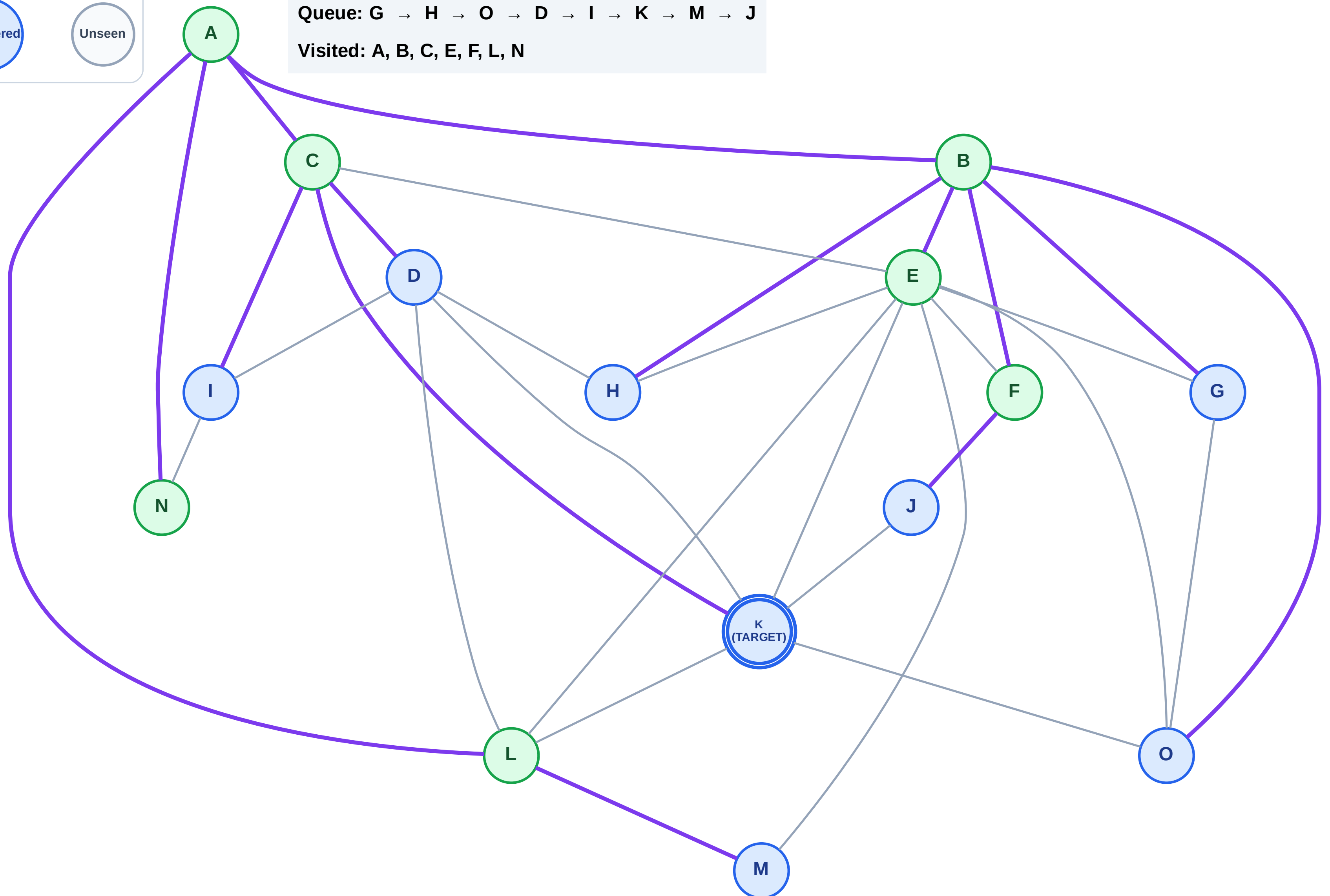
BFS Traversal

Step 15: Discover J from F



Queue: G → H → O → D → I → K → M → J

Visited: A, B, C, E, F, L, N



BFS Traversal

Step 16: Process node G

Legend

Complete

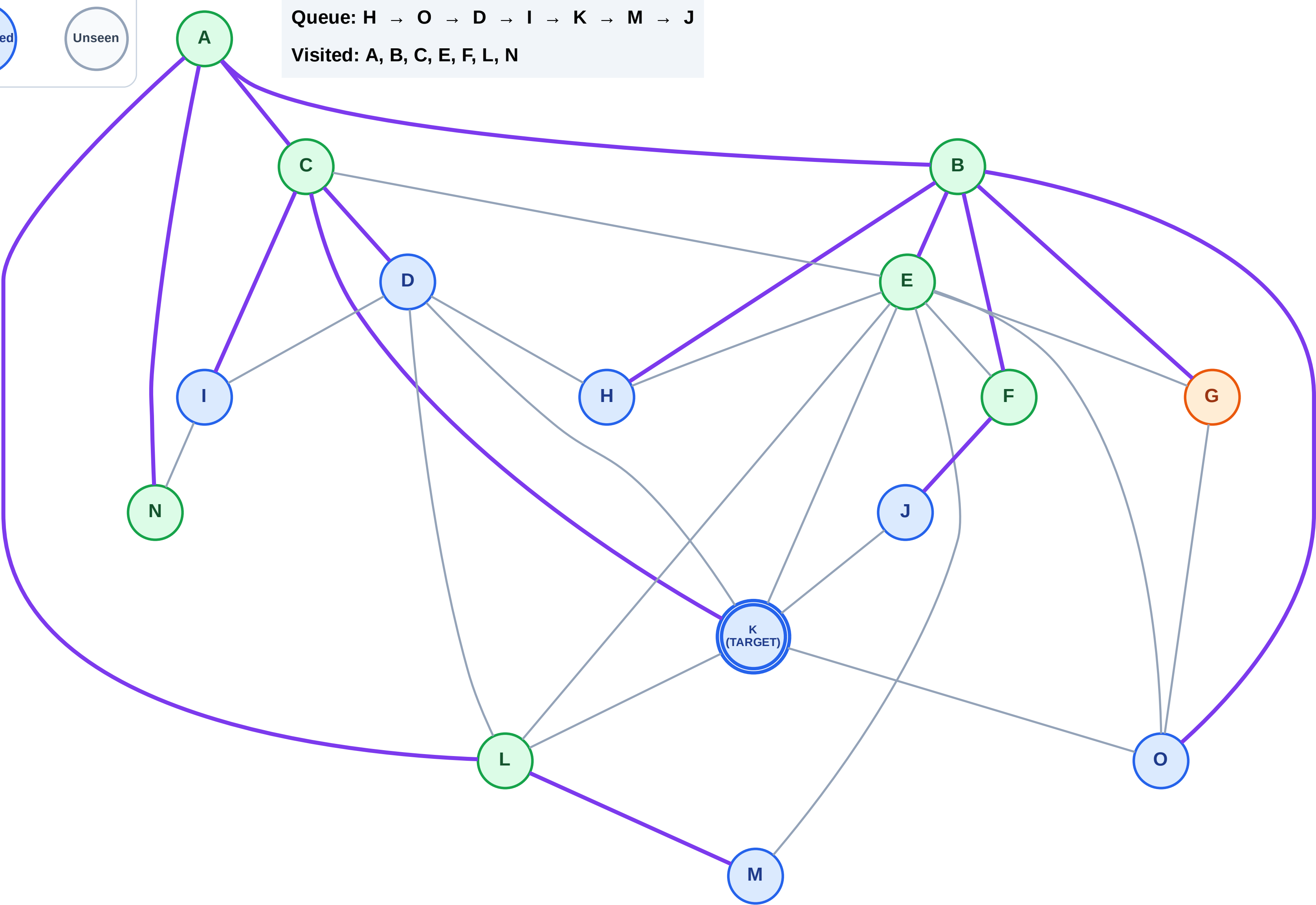
Processing

Discovered

Unseen

Queue: H → O → D → I → K → M → J

Visited: A, B, C, E, F, L, N



BFS Traversal

Step 17: No new neighbors from G

Legend

Complete

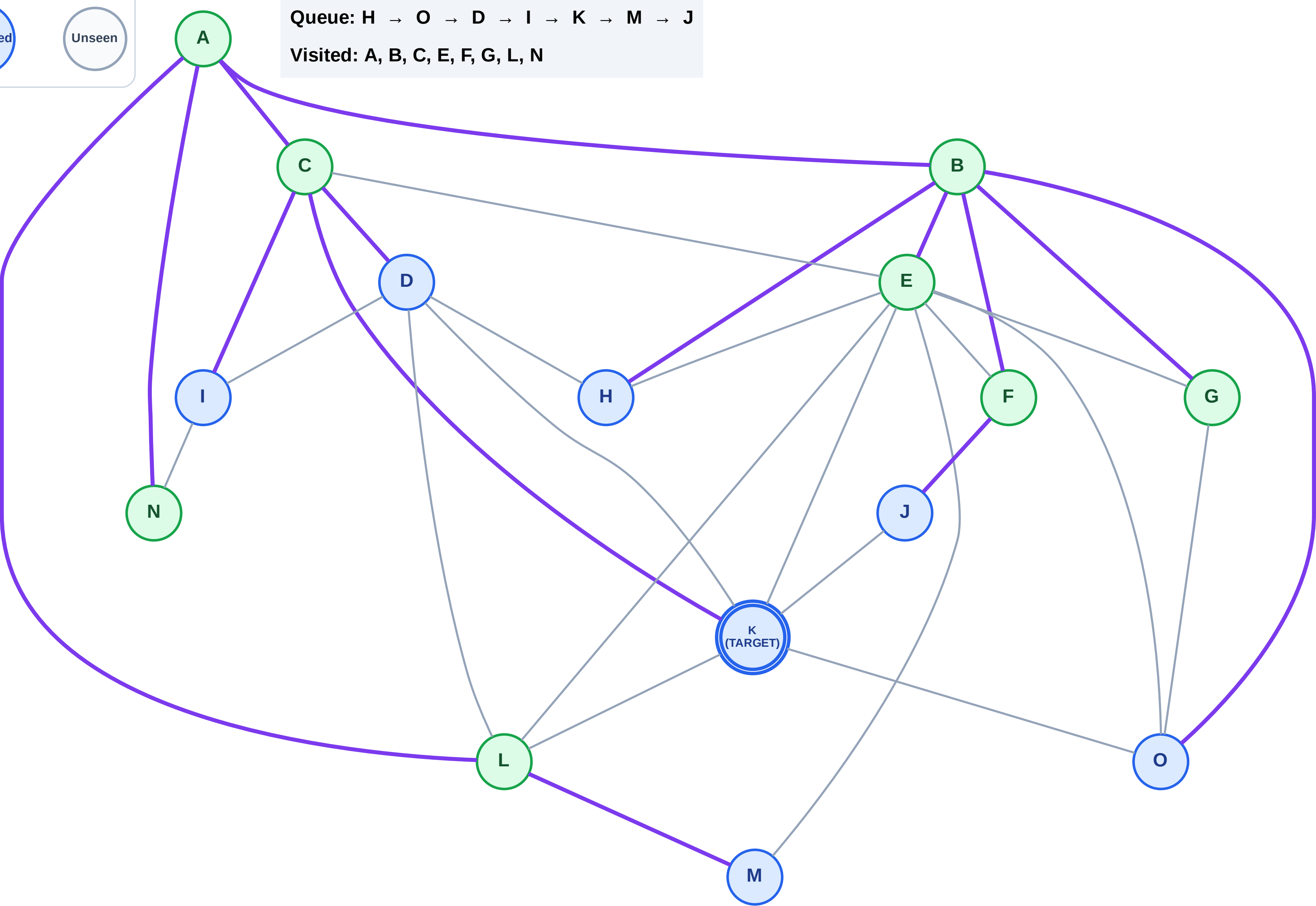
Processing

Discovered

Unseen

Queue: H → O → D → I → K → M → J

Visited: A, B, C, E, F, G, L, N



BFS Traversal

Step 18: Process node H

Legend

Complete

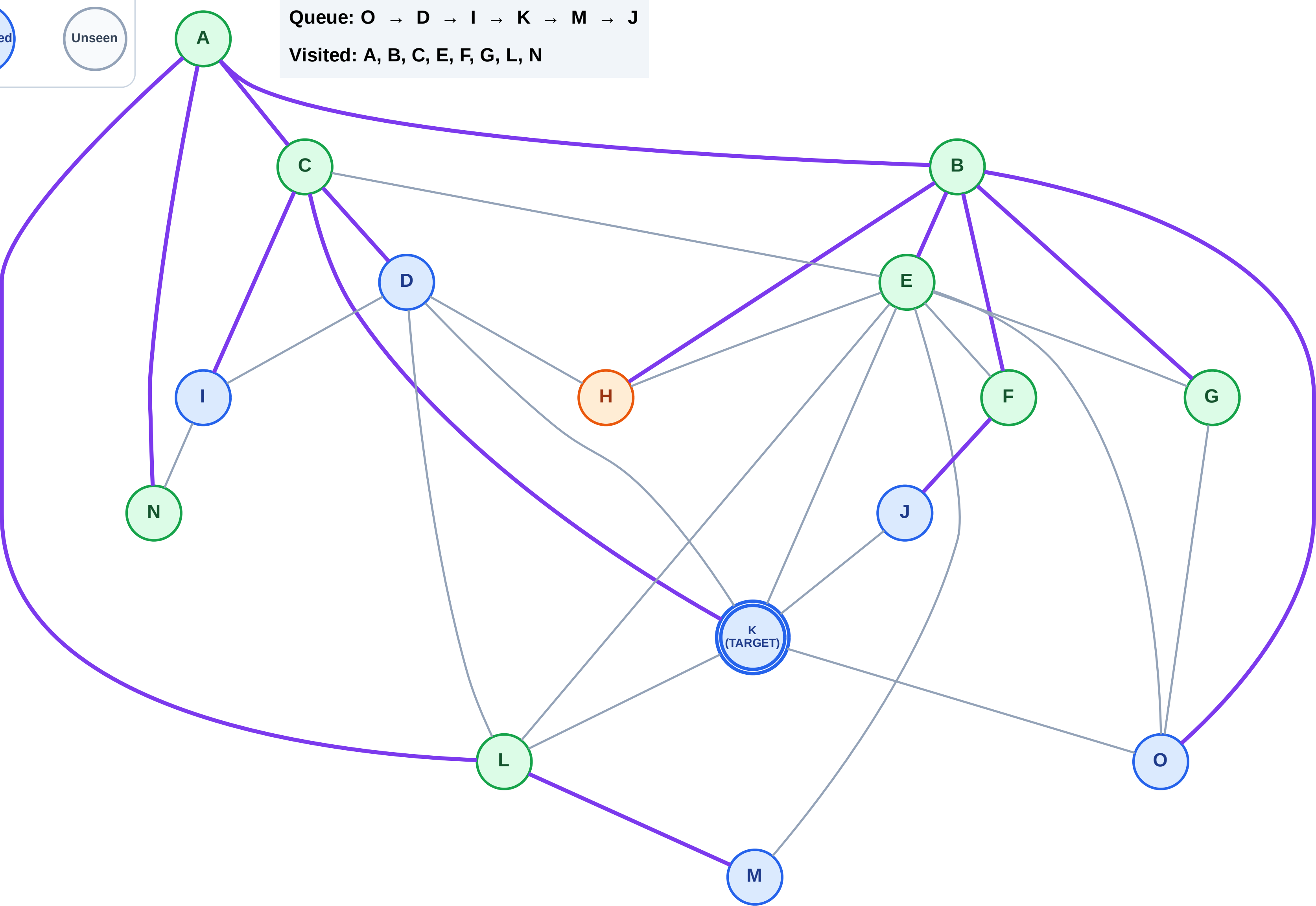
Processing

Discovered

Unseen

Queue: O → D → I → K → M → J

Visited: A, B, C, E, F, G, L, N



BFS Traversal

Step 19: No new neighbors from H

Legend

Complete

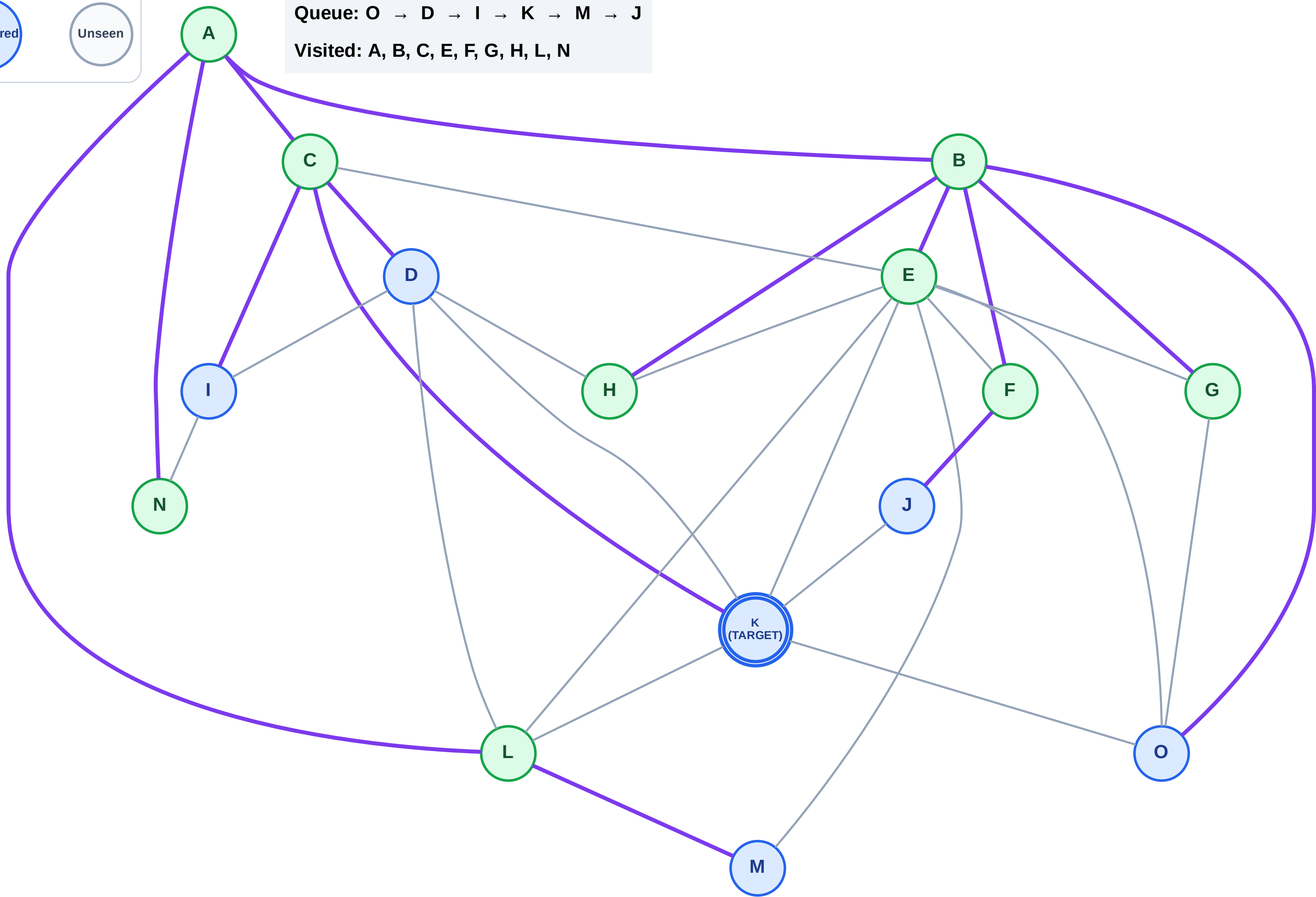
Processing

Discovered

Unseen

Queue: O → D → I → K → M → J

Visited: A, B, C, E, F, G, H, L, N



BFS Traversal

Step 20: Process node O

Legend

Complete

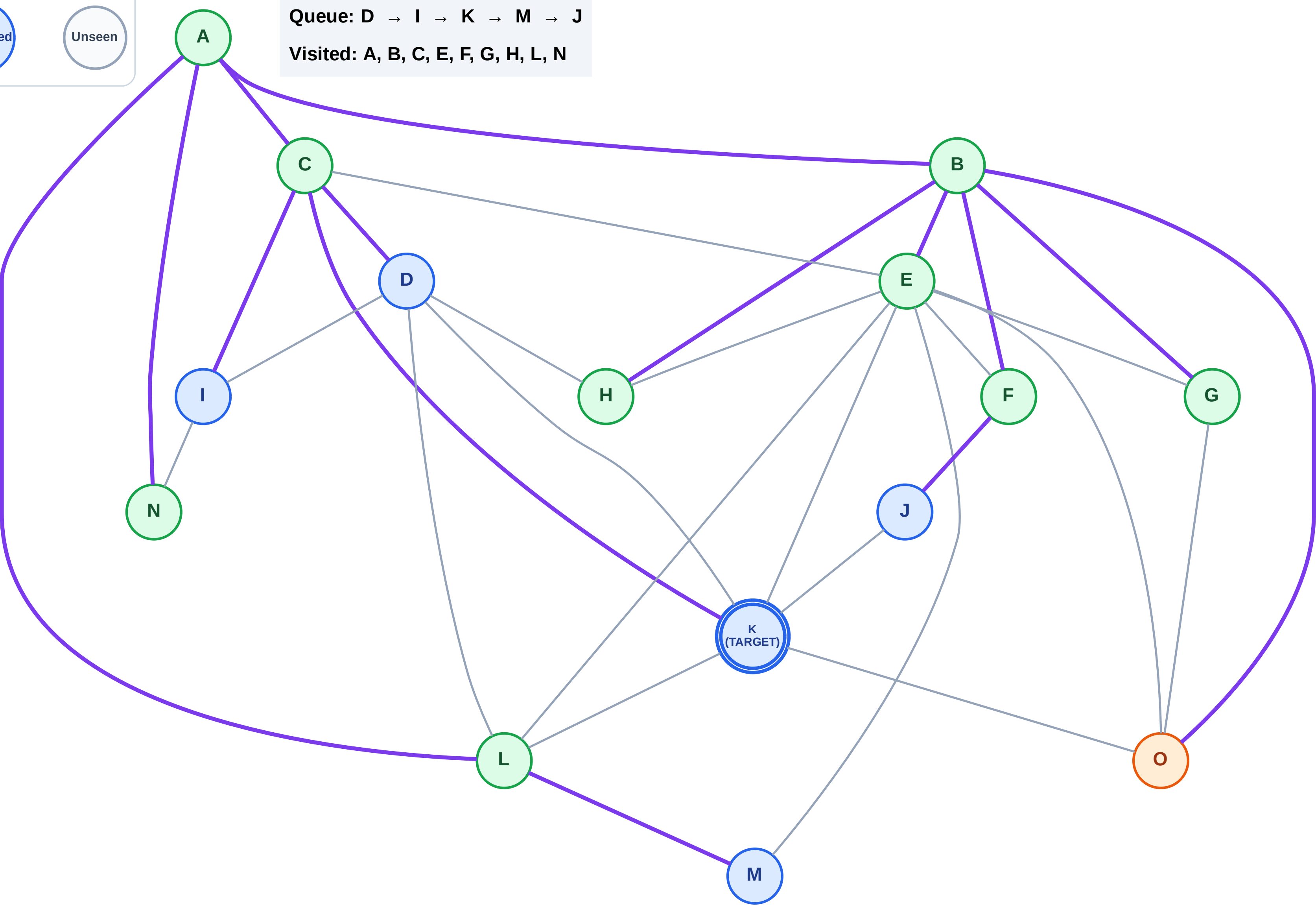
Processing

Discovered

Unseen

Queue: D → I → K → M → J

Visited: A, B, C, E, F, G, H, L, N



BFS Traversal

Step 21: No new neighbors from O

Legend

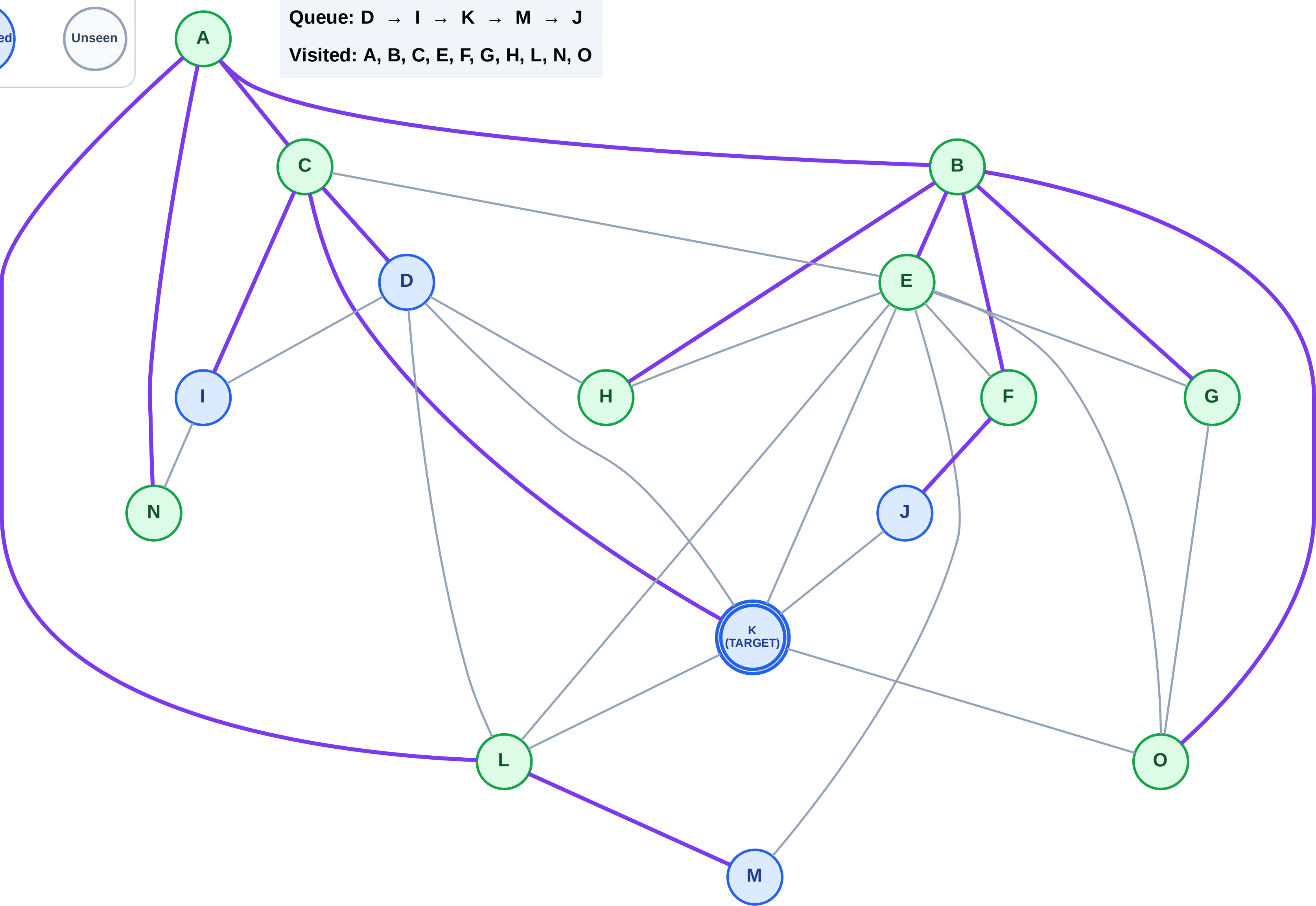
Complete

Processing

Discovered

Unseen

Queue: D → I → K → M → J
Visited: A, B, C, E, F, G, H, L, N, O



BFS Traversal

Step 22: Process node D

Legend

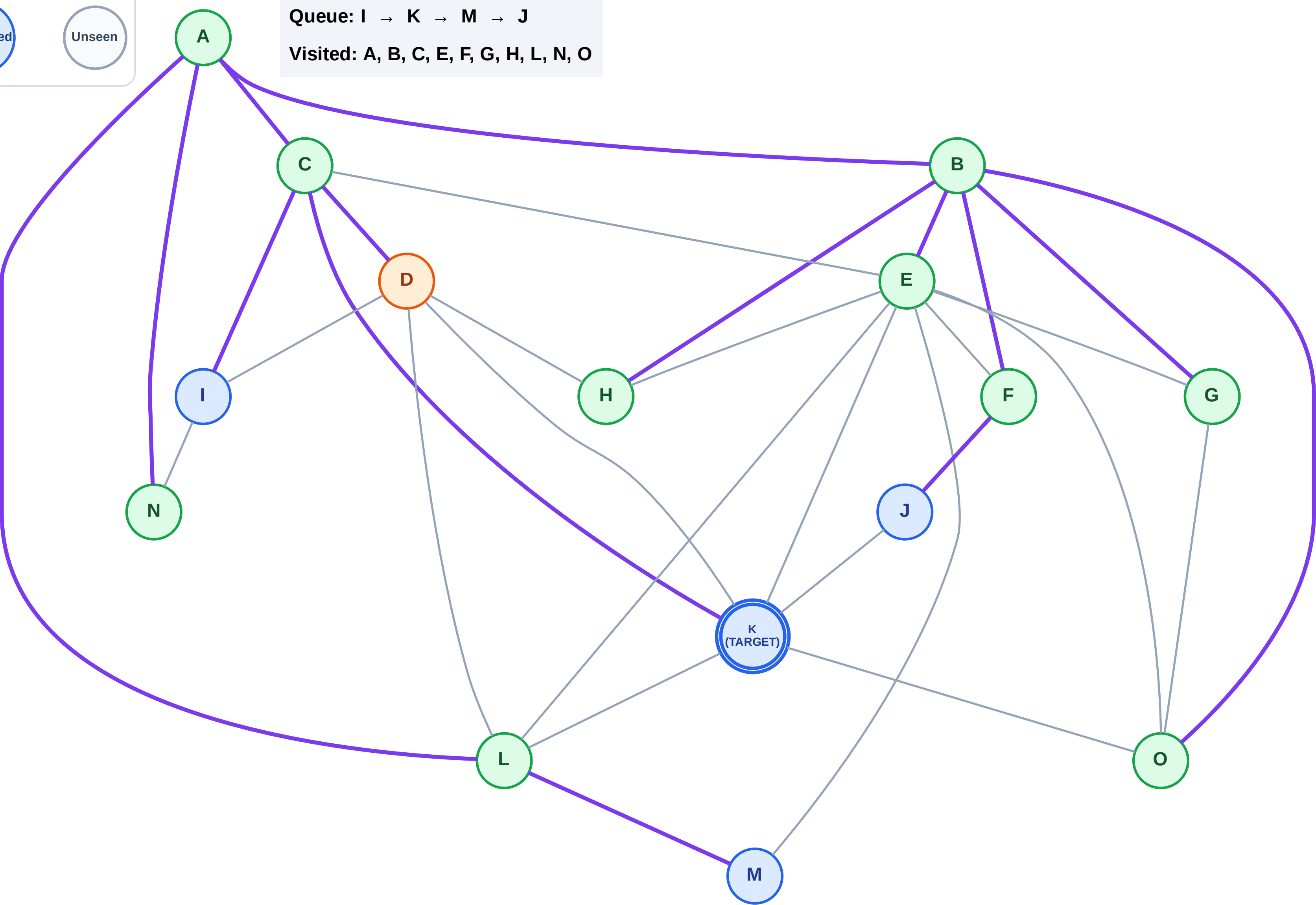
Complete

Processing

Discovered

Unseen

Queue: I → K → M → J
Visited: A, B, C, E, F, G, H, L, N, O



BFS Traversal

Step 23: No new neighbors from D

Legend

Complete

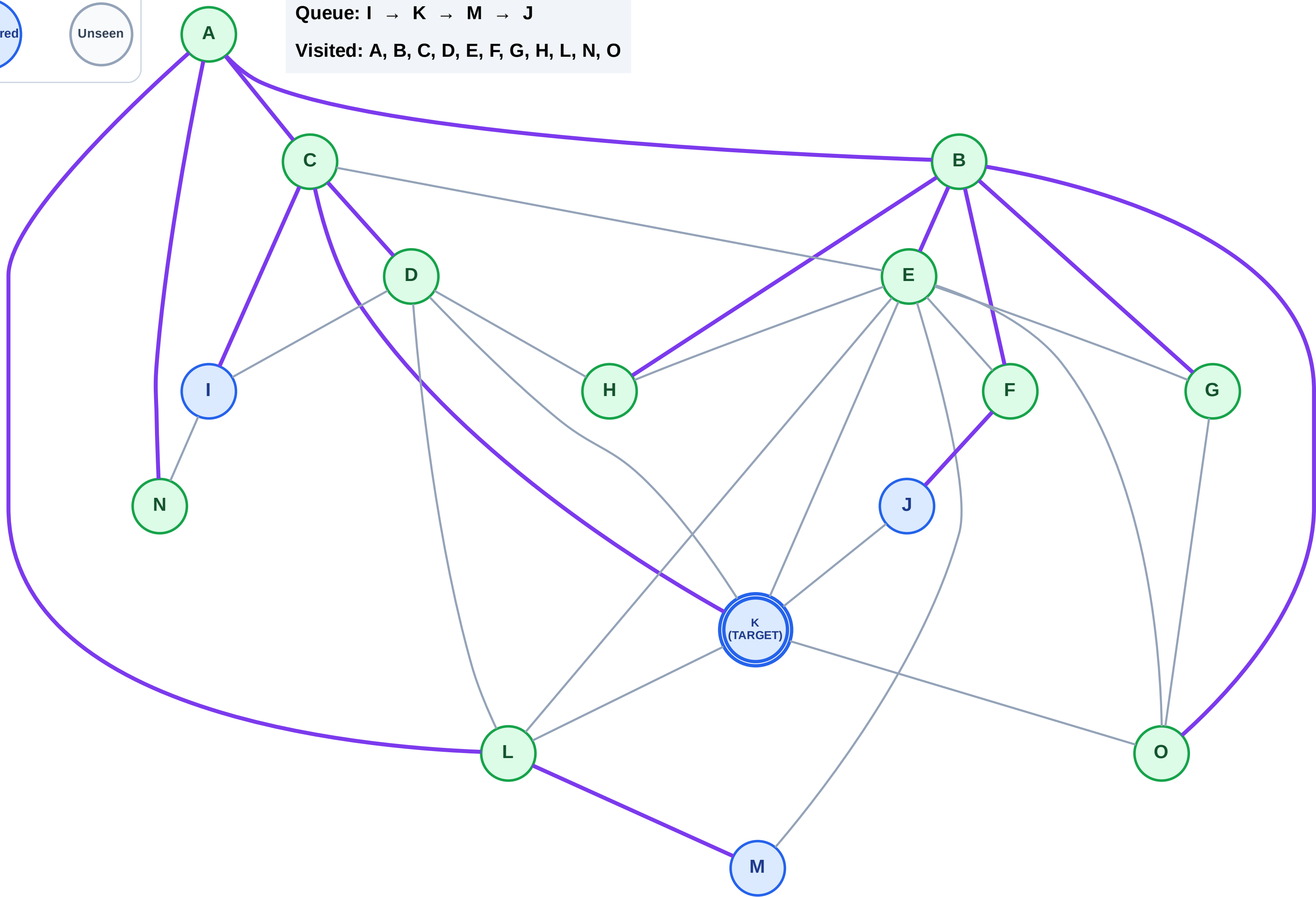
Processing

Discovered

Unseen

Queue: I → K → M → J

Visited: A, B, C, D, E, F, G, H, L, N, O



BFS Traversal

Step 24: Process node I

Legend

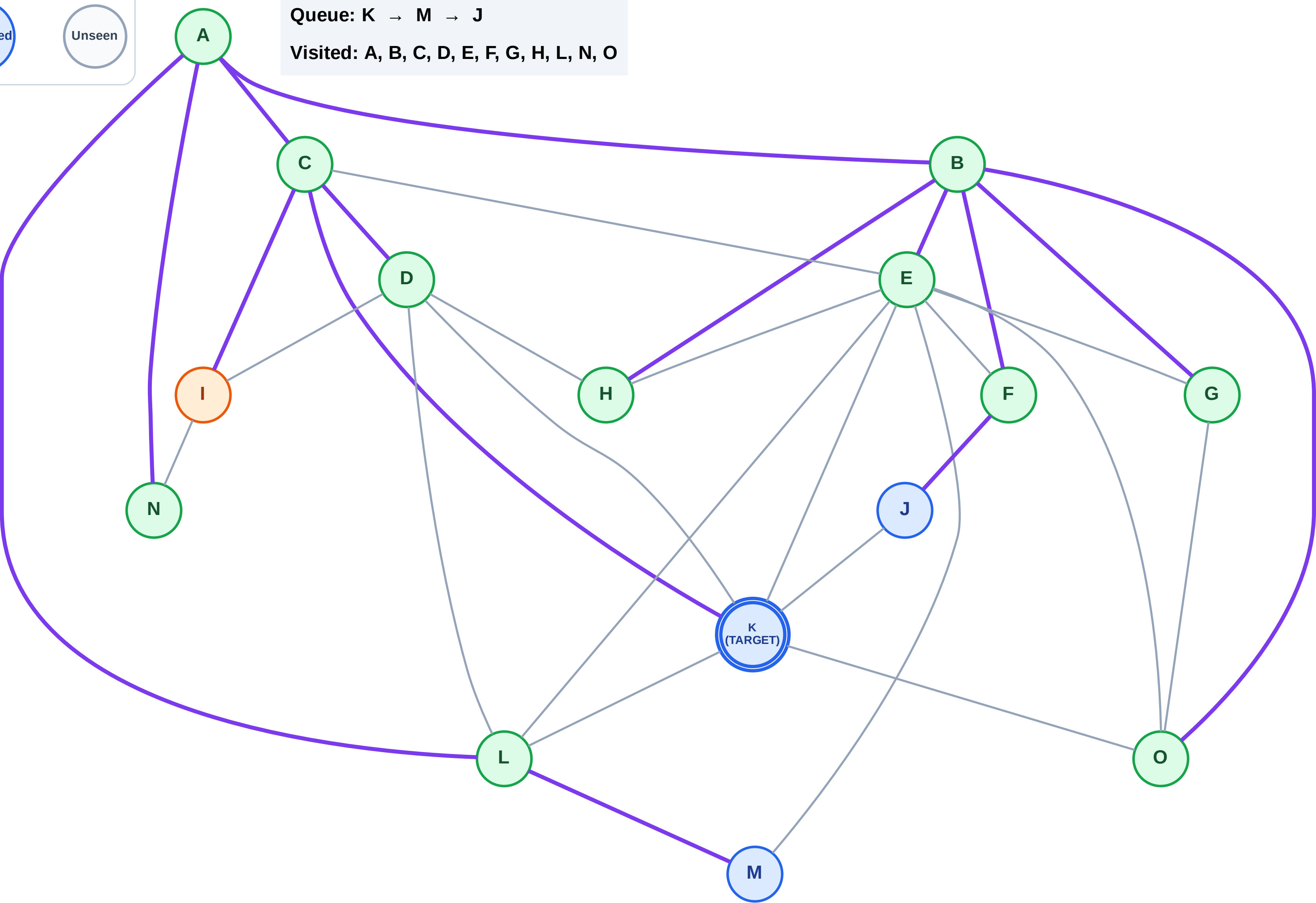
Complete

Processing

Discovered

Unseen

Queue: K → M → J
Visited: A, B, C, D, E, F, G, H, L, N, O



BFS Traversal

Step 25: No new neighbors from I

Legend

Complete

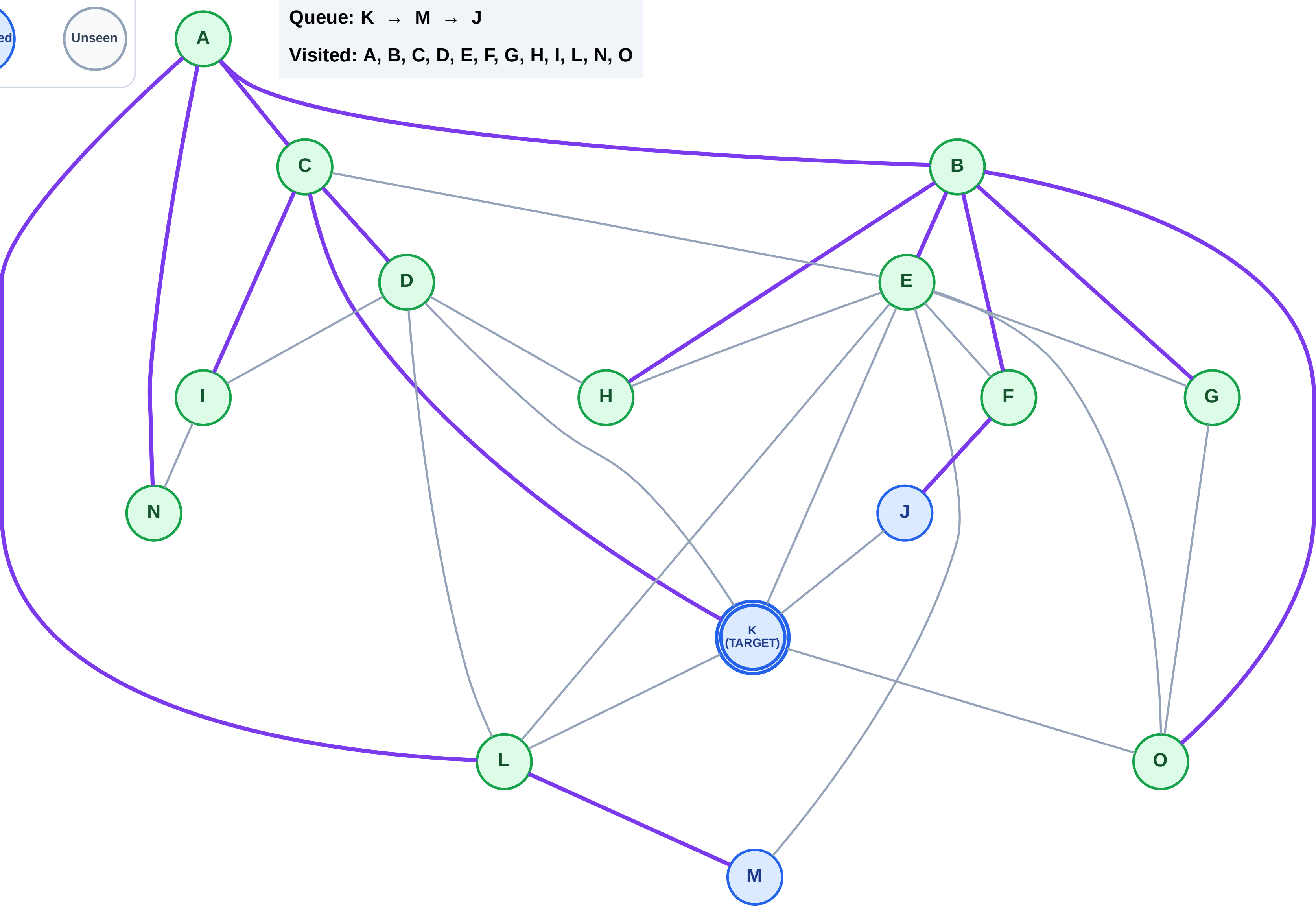
Processing

Discovered

Unseen

Queue: K → M → J

Visited: A, B, C, D, E, F, G, H, I, L, N, O



BFS Traversal

Step 26: Process node K

Legend

Complete

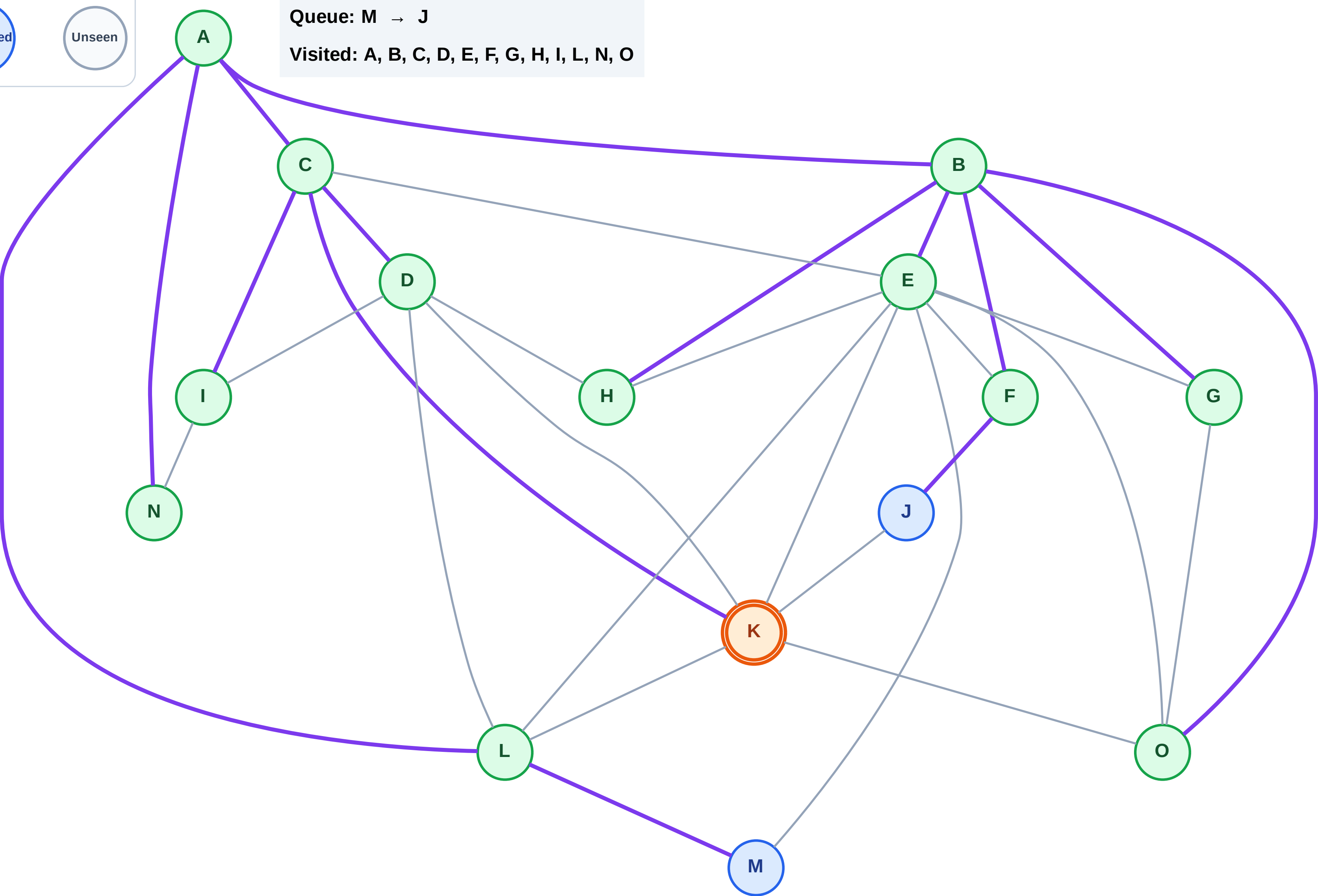
Processing

Discovered

Unseen

Queue: M → J

Visited: A, B, C, D, E, F, G, H, I, L, N, O



BFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

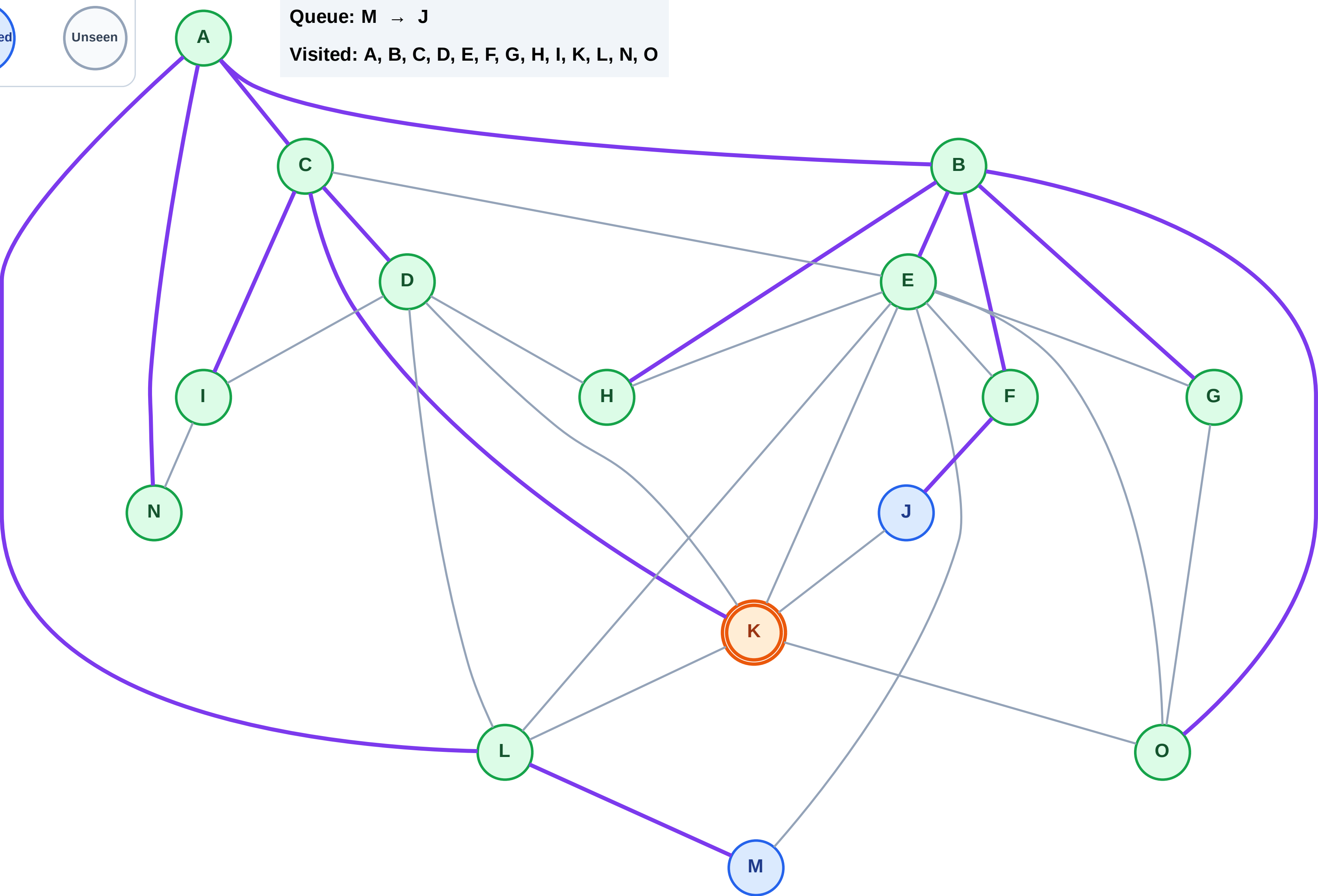
Processing

Discovered

Unseen

Queue: M → J

Visited: A, B, C, D, E, F, G, H, I, K, L, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

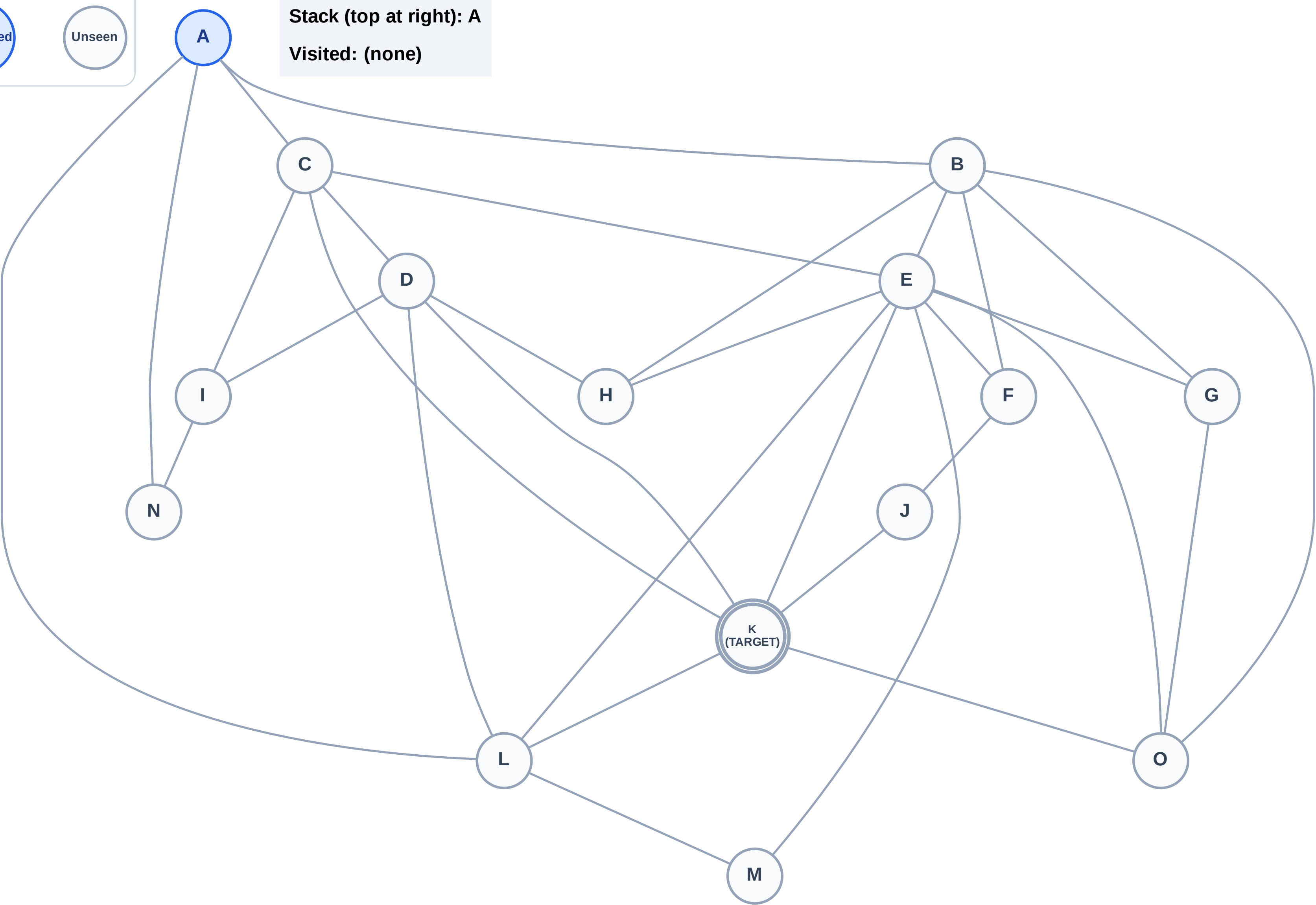
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

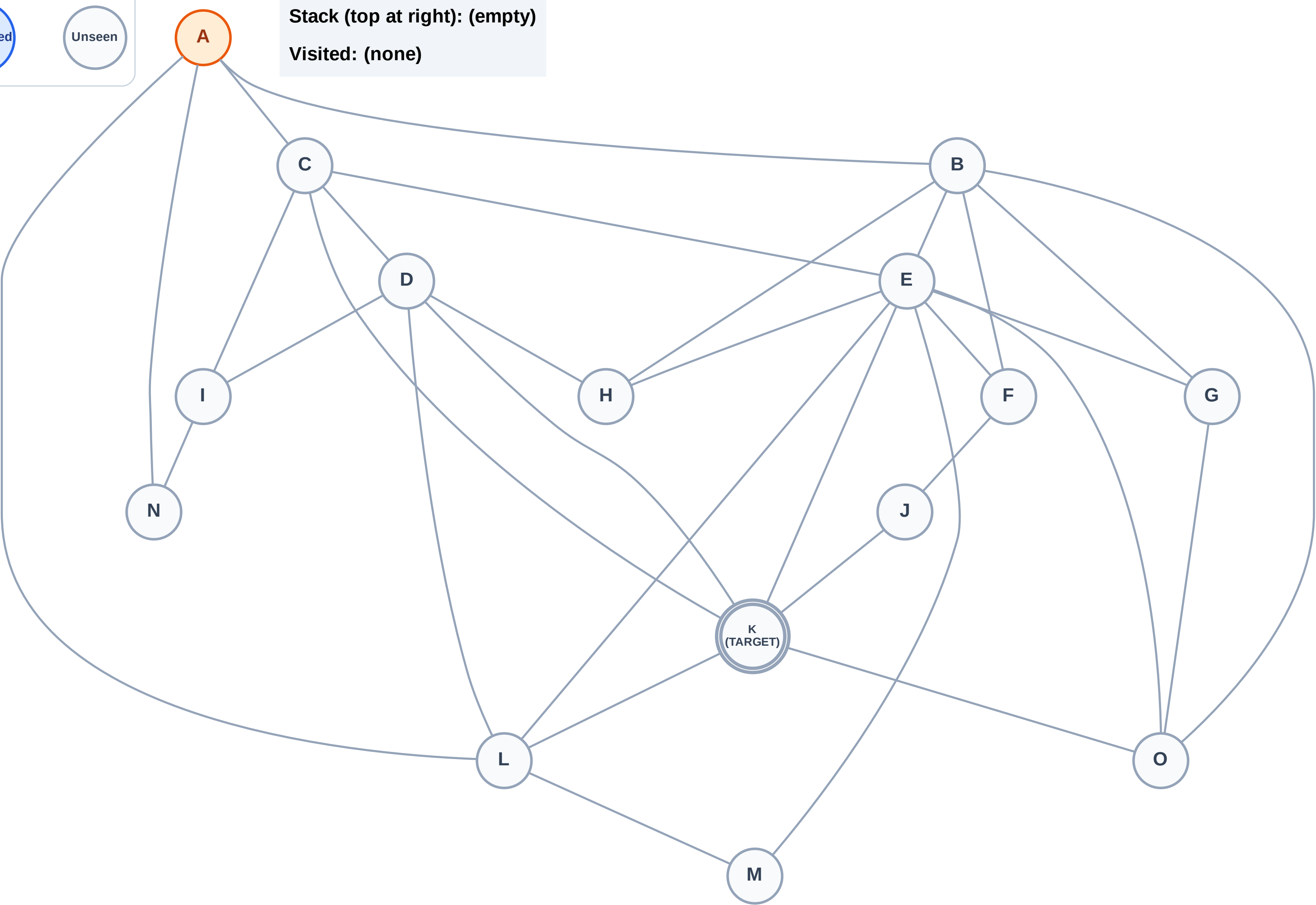
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover N, L, C, B from A

Legend

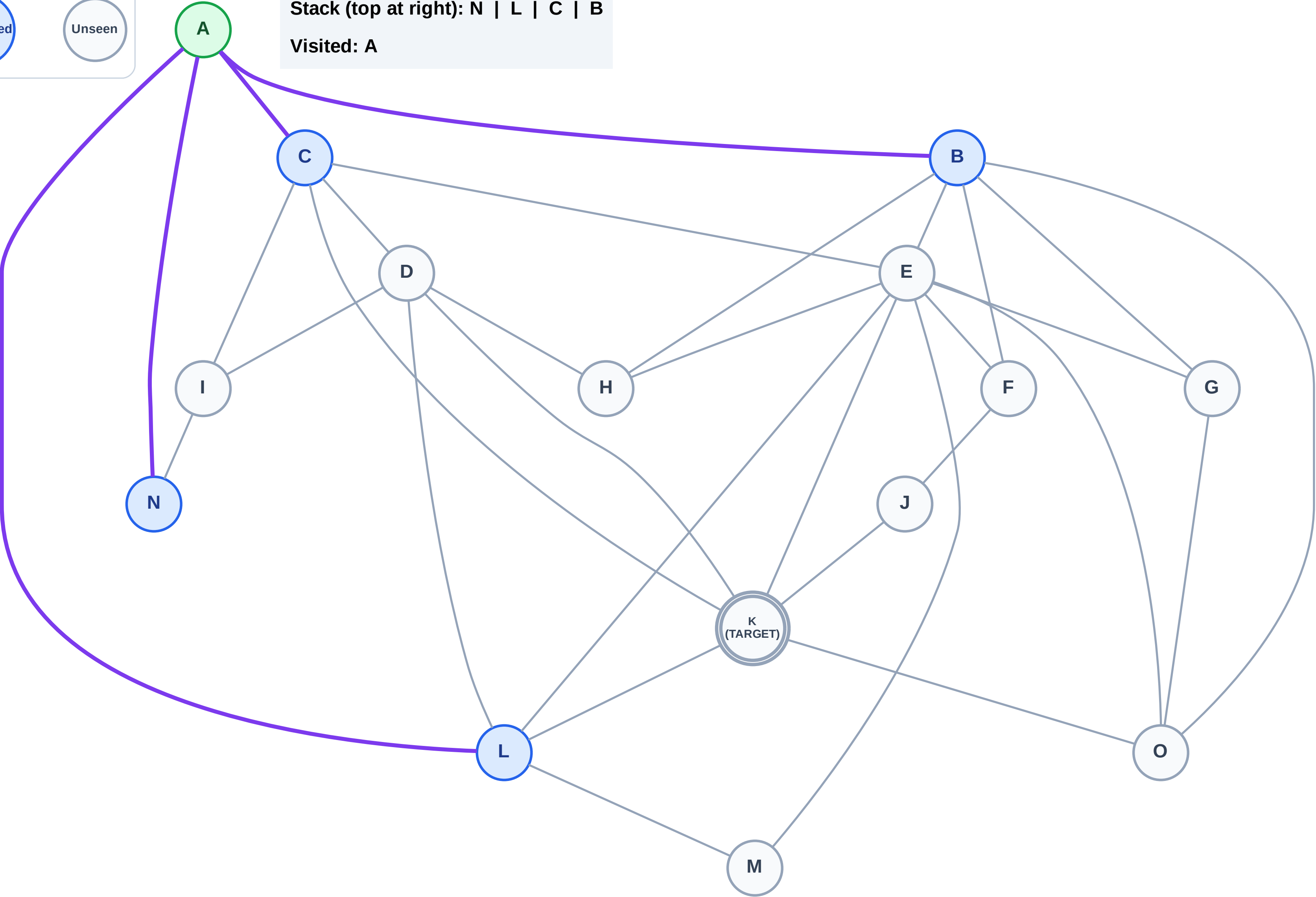
Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

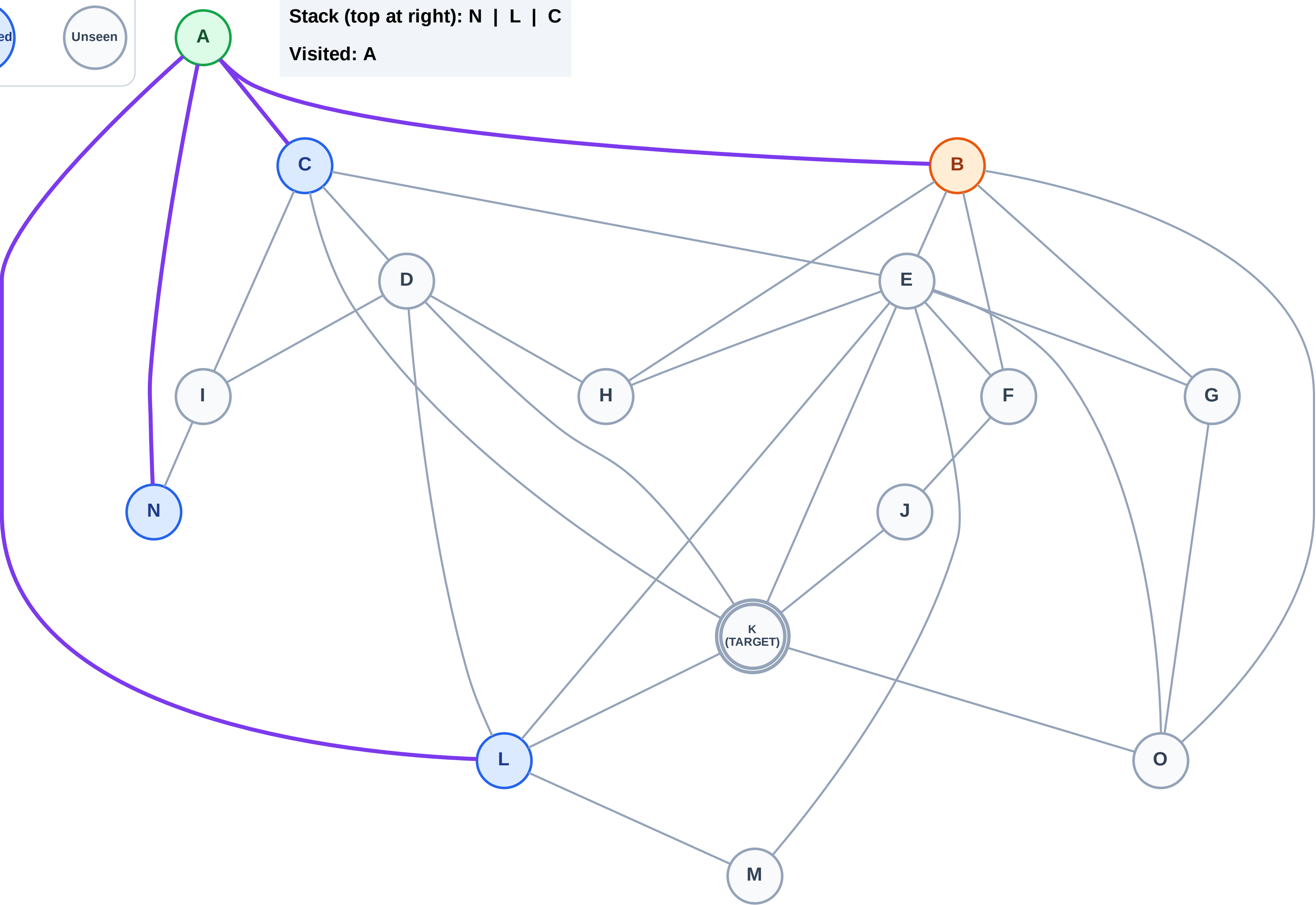
Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C
Visited: A



DFS Traversal

Step 5: Discover O, H, G, F, E from B

Legend

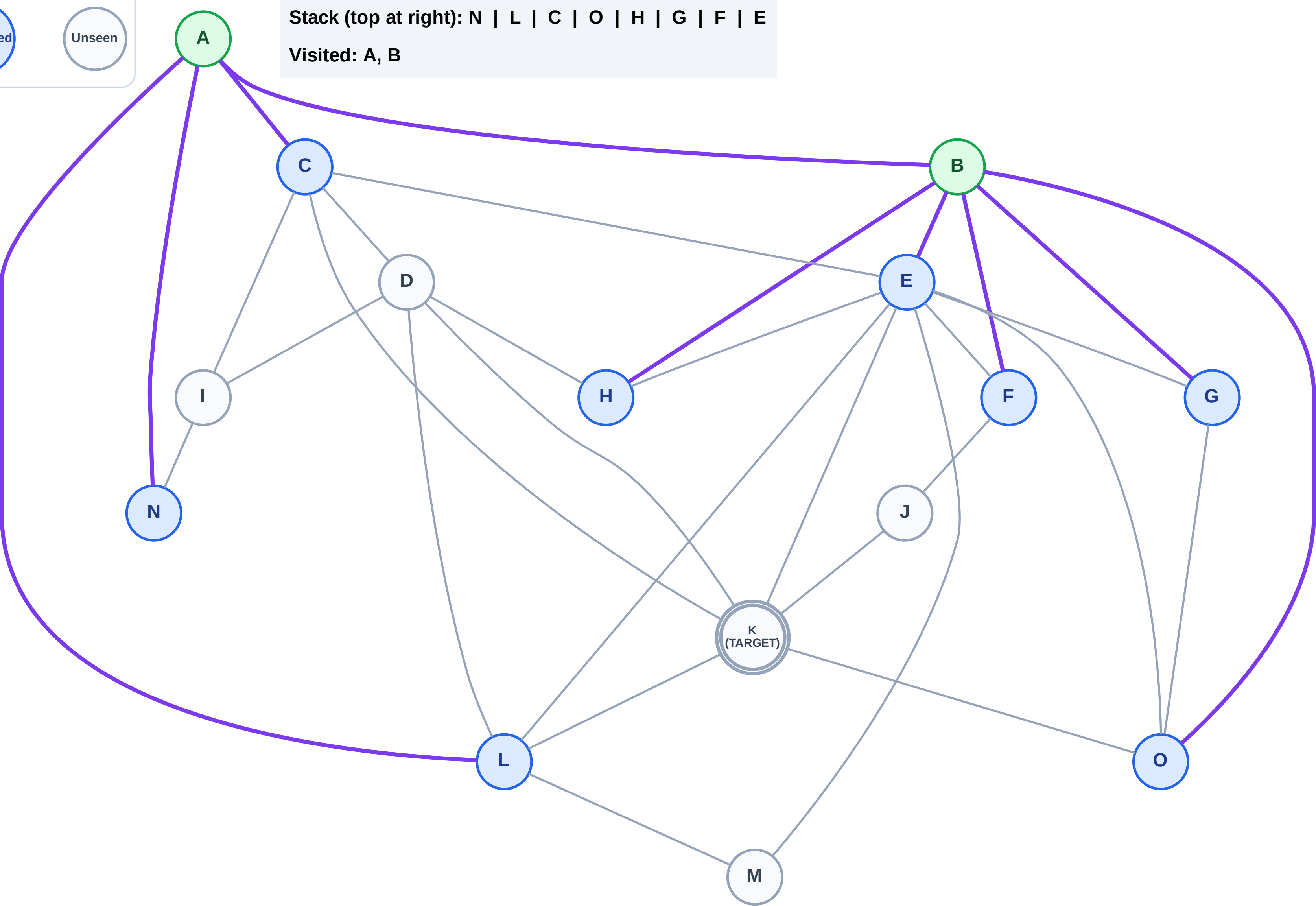
Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F | E
Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

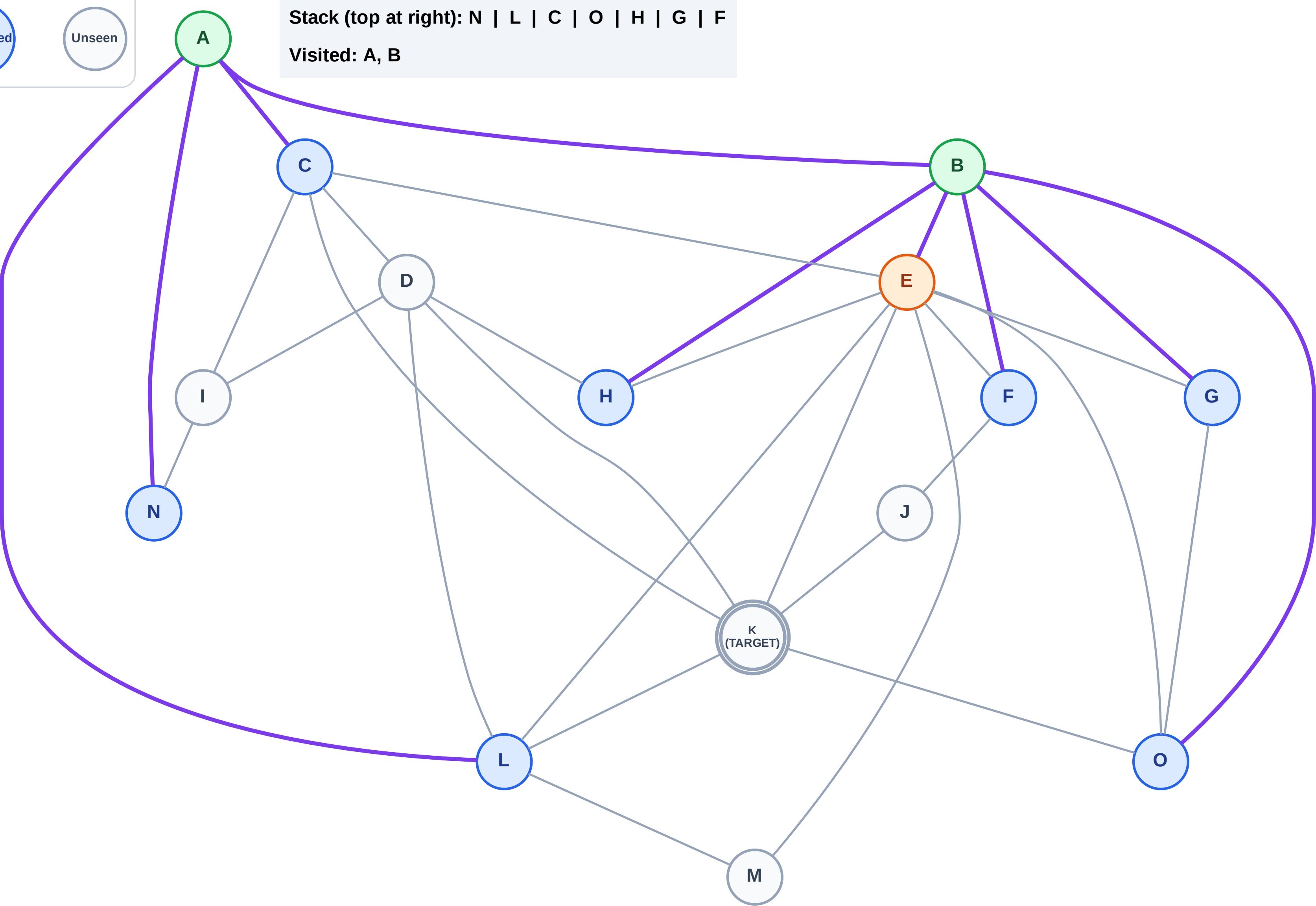
Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F

Visited: A, B



Step 7: Discover M, K from E

Step 7: Discover M, K from E

Complete

Complete



Processing

Processing



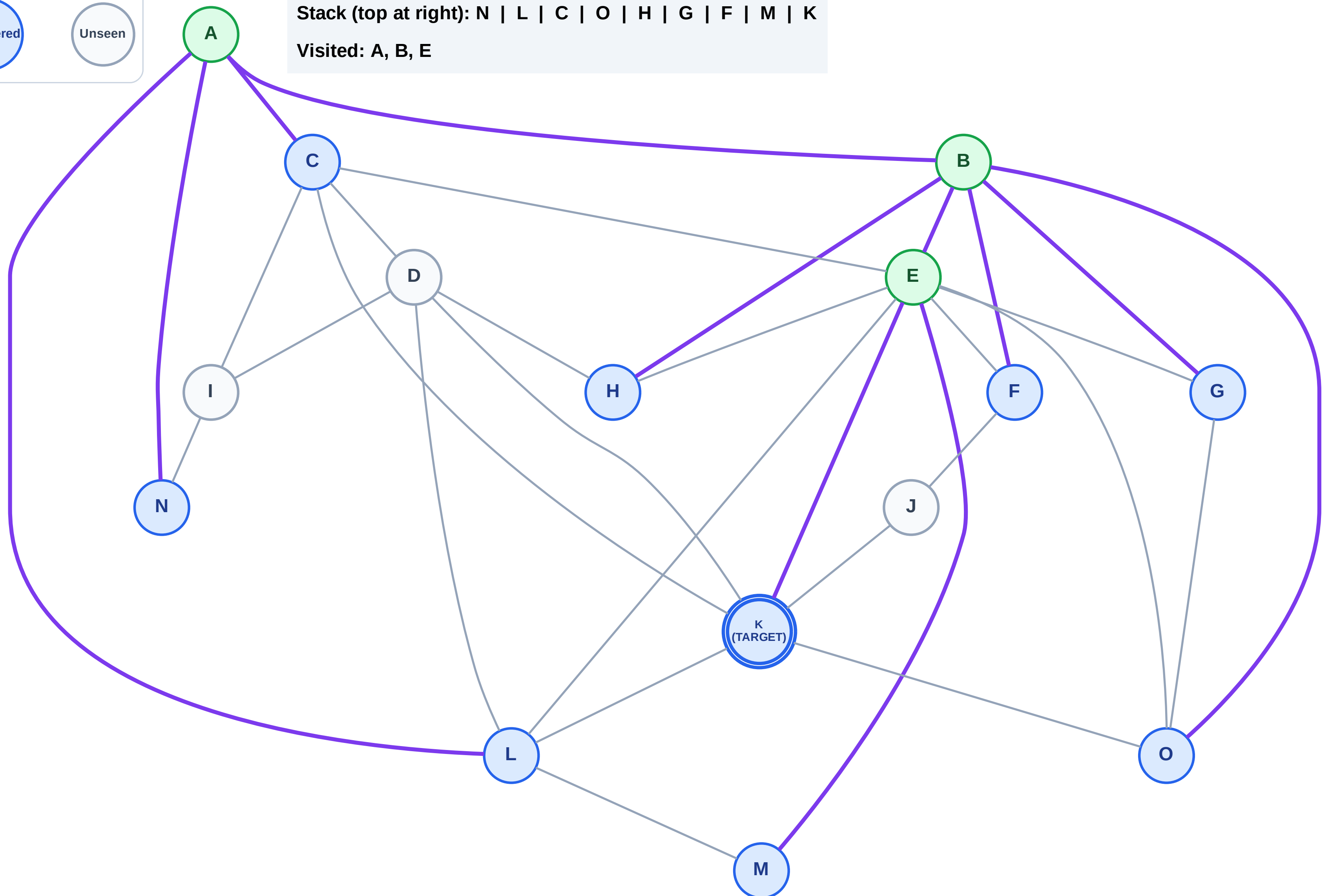
Discovered

Discovered



Unseen

Visited: A, B, E



DFS Traversal

Step 8: Process node K

Legend

Complete

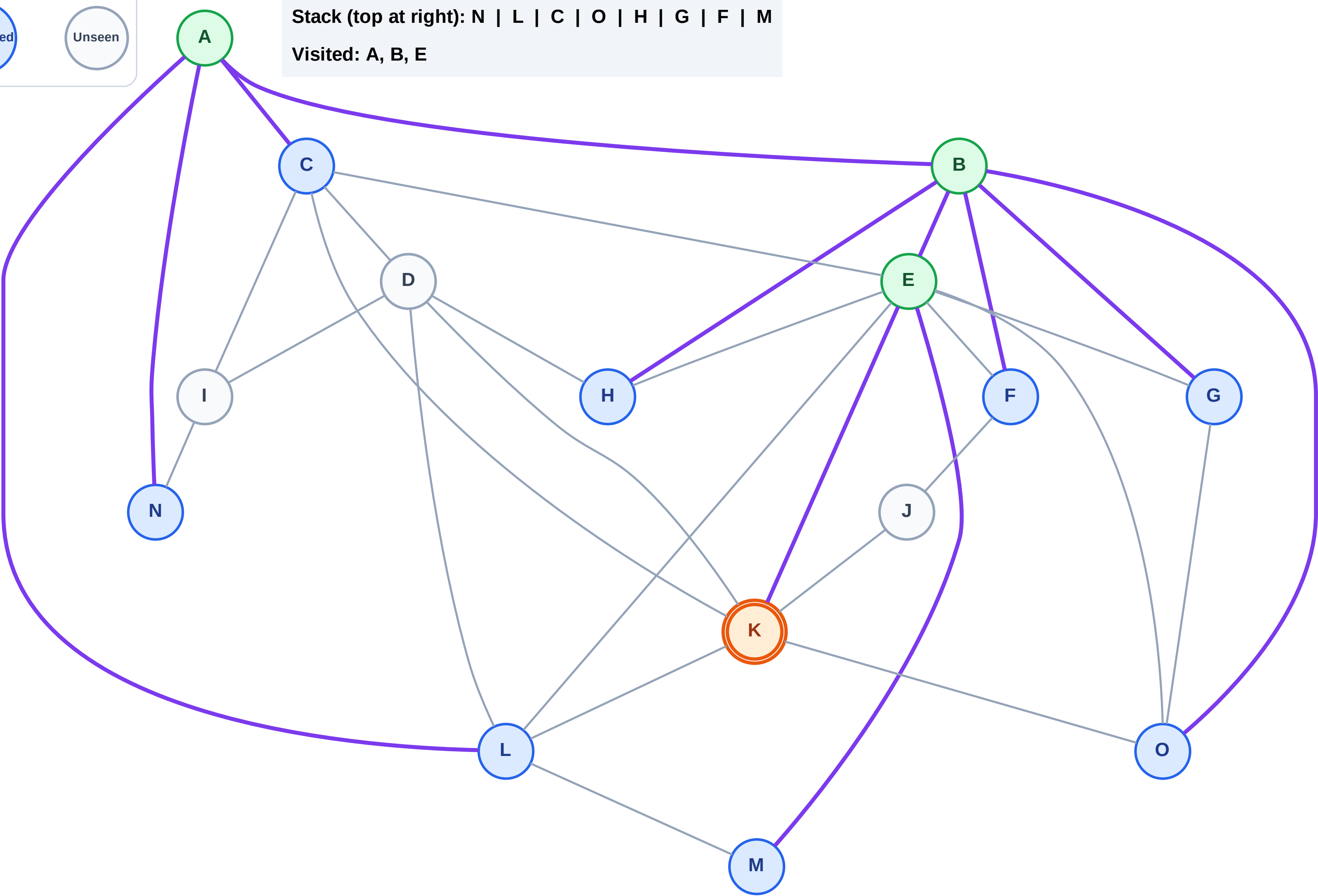
Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F | M

Visited: A, B, E



DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F | M

Visited: A, B, E, K

